# Mbarara Integrated Concrete Products Factory

Prompt Coverage, Module Specification & Integration Supplement

This supplement was added after review of the merged PDF to explicitly capture every prior prompt and requirement in one downloadable document. It is prepended to the full detailed investor report and recent module master report.

Currency: UGX only. Market and competitor values marked Estimated, Quotation Required, or Needs verification must be re-verified before investment decisions.

# Prompt Coverage Matrix

The table below maps each user request to the corrected merged report and platform deliverables.

| User request / prompt | Covered in corrected merged PDF | Where it appears |
|---|---|---|
| Professional 150-200 page investor-grade PDF report generator for the Mbarara Integrated Concrete Products Factory, UGX only. | Yes | Detailed investor report section appended after this supplement; full report is 194 pages and includes A4 portrait consultancy-report layout, table of contents, chapters, tables, charts, diagrams, image placeholders, and appendices. |
| Financial Simulator using React, Tailwind CSS, Recharts, local editable state, UGX inputs, product volumes/prices, cost/profit outputs, charts, CSV and PDF export. | Yes | Platform Financials tab, Product Costing tables, Financial Model chapter, ERP financial summary, and prompt coverage module specification. |
| Inventory Management System tracking cement, stone dust, sand, aggregates, diesel, water, pallets, moulds, spare parts; add/issue stock; low-stock alerts; supplier/truck/cost; valuation; monthly usage; reorder calculation. | Yes | Platform Inventory tab, PostgreSQL inventory and inventory_transactions tables, Raw Material Consumption chapter, Inventory module specification. |
| Quality Control Database tracking every production batch, material usage, mix ratio, curing, tests, pass/fail outputs, rejection rate, strength by product, defect trends, printable certificate. | Yes | Platform QC Lab tab, Production Batch register, quality_tests schema, Laboratory and Quality Control chapter, QC module specification. |
| Simple ERP system with Sales, Inventory, Production, Quality Control, Customers, Suppliers, Expenses, Reports; sales/production/report fields. | Yes | ERP tab, Database schema, ERP Integration Summary chapter, customer/supplier balances, daily sales, monthly profit, inventory usage, production efficiency. |
| Continue: next module integrated with existing ERP. | Yes | Dispatch & Delivery was added in the previous ERP iteration and documented here as part of ERP/dispatch integration history and operating workflow. |
| React + TypeScript + Tailwind CSS + Recharts, modular architecture, maintainability and scalability. | Yes | Platform architecture section, source files, components/common, layout, data, db, lib, modules structure. |
| PostgreSQL database structure for products, suppliers, customers, production batches, inventory, sales, expenses, quality tests; future ERP integration. | Yes | PostgreSQL tab and src/db/postgres-schema.sql; schema summary and relationship matrix in this supplement. |
| Inventory Management Module connected to Financial Simulator. | Yes | Shared AppState and computeErp/computeFinancials/computeInventory calculations; material prices and stock values feed financial reporting. |
| Quality Control and Laboratory Module integrated with production batches. | Yes | Batch IDs link production_batches to quality_tests; certificate uses batch/product/operator/approval data. |
| ERP module connected to sales, inventory, production, quality control, expenses. | Yes | ERP tab and PostgreSQL relationship matrix; reports combine sales, balances, stock usage, production efficiency, QC pass rate, expenses. |
| Competitor and Market Intelligence Database module with competitors, price intelligence, customer demand, gap scoring, SWOT, dashboard, focus areas, seed competitors, source tracking, CSV/PDF exports. | Yes | Market Intel tab, Market Analysis and Competitor Analysis chapters, Market Gap Analysis chapter, source-confidence notes, and export specification. |
| Reusable ReportPdfGenerator.tsx with jsPDF/pdfmake, page numbers, headings, tables, Recharts images, confidence labels, assumption labels, disclaimer, Download Master PDF button. | Yes | ReportPdfGenerator.tsx, Master PDF tab, saved downloadable PDF link, and this corrected merged PDF. |

# Module Specifications Included

## Financial Simulator

| Area | Specification |
|---|---|
| Inputs | Cement price per 50 kg bag; stone dust, sand, aggregate per tonne; electricity per kWh; water per m3; diesel per litre; labour per month; transport per truck; working days; daily production; selling price per product. |
| Products | 6-inch hollow blocks, 8-inch hollow blocks, solid blocks, 60 mm pavers, 80 mm pavers, kerbstones, drainage channels, culverts; broader portfolio also includes 4-inch blocks, grass/permeable pavers, cobblestone pavers, ready-mix concrete. |
| Outputs | Cost per product, monthly revenue, material cost, utility cost, labour cost, gross profit, net profit, break-even point, ROI estimate, and whether UGX 20,000,000/month is reached. |
| Charts | Revenue by product, profit by product, cost breakdown, monthly profit projection. |
| Exports | CSV export and PDF summary export in the market/financial reporting flow; master PDF consolidates final report. |
| Integration | Inventory prices and ERP sales/expenses share the same UGX model and calculations. |

## Inventory Management

| Area | Specification |
|---|---|
| Tracked items | Cement bags, stone dust tonnes, sand tonnes, aggregates tonnes, diesel litres, water m3, pallets, moulds, spare parts. |
| Transactions | Add stock and issue stock to production with supplier, delivery truck number, cost per delivery, production batch link, and notes. |
| Controls | Low-stock alerts, stock valuation in UGX, daily consumption, days remaining, monthly usage report, reorder level calculation. |
| Dashboard | Current inventory, stock value, daily consumption value, days of stock remaining, top consumed materials. |
| Financial connection | Unit costs and stock values support financial cost assumptions and ERP material usage reporting. |

## Quality Control and Laboratory

| Area | Specification |
|---|---|
| Batch fields | Batch ID, date, product type, operator, cement used, stone dust used, sand used, water used, mix ratio, curing start date, curing age, test result, approval status. |
| Tests | Compressive strength, water absorption, density, dimensions, visual defects. |
| Outputs | Passed batches, failed batches, rejection rate, average strength by product, defect trends, printable test certificate. |
| Production integration | QC tests are linked to production batches by batch ID and product ID. |

## ERP

| Area | Specification |
|---|---|
| Modules | Sales, Inventory, Production, Quality Control, Customers, Suppliers, Expenses, Reports; Dispatch & Delivery integration from the continuation step. |
| Sales | Customer name, product, quantity, unit price, delivery cost, paid amount, balance. |
| Production | Daily production, product type, material consumption, machine hours, rejected products. |
| Reports | Daily sales, monthly profit, inventory usage, production efficiency, customer balances, supplier balances, QC pass rate. |
| Integration | All modules share local editable state and are structured for PostgreSQL migration. |

## Market Intelligence

| Area | Specification |
|---|---|
| Competitors | Competitor ID, name, location, district, region, GPS if available, business type, products, prices, capacity, machine type, delivery radius, contact, links, strengths, weaknesses, threat level, notes, source URL, last updated. |

| Area | Specification |
|---|---|
| Price intelligence | Tracks market prices for blocks, pavers, kerbstones, drainage channels, culverts, ready-mix concrete with supplier, location, unit, UGX price, delivery inclusion, date checked, source URL, confidence. |
| Customer demand | Contractors, hardware stores, property developers, schools, hospitals, churches, NGOs, government projects, fuel stations, hotels, residential builders with likely products, demand, buying power, payment reliability. |
| Scoring | Opportunity score = demand + margin + strategic value - competition - logistics difficulty. |
| SWOT | Automatic strengths, weaknesses, opportunities, threats, and response strategy for each competitor. |
| Dashboards | Competitor count by district, product prices by supplier, average market price, opportunity score, threat distribution, target customer distribution, planned vs competitor price. |
| Focus areas | Mbarara City, Rwampara, Isingiro, Ntungamo, Bushenyi, Ibanda, Kiruhura, Sheema, Lyantonde, Kabale, Rukungiri. |
| Seed entries | iConic Concrete, Master Industries, Stone Providers Ltd, Nicomah, BIMS Bloks Mbarara, local block yards, local quarry suppliers, Hima/Tororo/Kampala/Simba Cement distributors, all marked Needs verification. |

## PDF Master Report

| Area | Specification |
|---|---|
| Required chapters | Cover page, table of contents, executive summary, market analysis, competitor analysis, product portfolio, mix ratios, raw material consumption, machinery, factory layout, utilities, QC, legal roadmap, financial model, risk register, implementation roadmap, East African expansion, appendices. |
| Format | A4 portrait consultancy report, not slide style; chapters, paragraphs, tables, figures, appendices, page numbers. |
| Data confidence | Verified / Estimated / Quotation Required / Needs verification labels and assumption notes. |
| Disclaimer | Market prices and competitor data must be re-verified before investment decisions. |
| Download | Direct app URL serves the corrected full merged PDF. |

# PostgreSQL Integration Structure

The platform includes a PostgreSQL-ready schema designed for future ERP integration. Monetary fields are UGX-only, and market intelligence is separated from accounting/ERP tables so unverified records can be governed separately.

| Table | ERP purpose |
| --- | --- |
| products | Product master, portfolio, planned UGX prices, mix ratios, confidence labels. |
| suppliers | Supplier master for cement, quarry, diesel, water, spares, logistics. |
| customers | Customer demand, customer type, target products, payment reliability. |
| production_batches | Batch production, operators, product, material consumption, rejects, curing. |
| inventory | Stock master, opening/current quantity, unit costs, reorder levels, daily use. |
| inventory_transactions | Add stock and issue-to-production records linked to suppliers and batches. |
| sales | Sales records linked to customers and products with UGX price, delivery, paid amount. |
| expenses | Operating cost records linked to supplier/payee where applicable. |
| quality_tests | Batch-linked lab tests, strength, absorption, density, dimensions, visual defects, approval. |
| competitor_records | Market intelligence extension with source tracking and verification status. |

# Source Files and Download Locations

| Artifact | Path / role |
| --- | --- |
| React platform | outputs/mbarara-concrete-factory-platform/src/App.tsx |
| Reusable PDF generator | outputs/mbarara-concrete-factory-platform/src/modules/ReportPdfGenerator.tsx |
| Financial module | outputs/mbarara-concrete-factory-platform/src/modules/FinancialModule.tsx |
| Inventory module | outputs/mbarara-concrete-factory-platform/src/modules/InventoryModule.tsx |
| Production module | outputs/mbarara-concrete-factory-platform/src/modules/ProductionModule.tsx |
| Quality module | outputs/mbarara-concrete-factory-platform/src/modules/QualityModule.tsx |
| ERP module | outputs/mbarara-concrete-factory-platform/src/modules/ErpModule.tsx |
| Market intelligence module | outputs/mbarara-concrete-factory-platform/src/modules/MarketIntelligenceModule.tsx |
| PostgreSQL schema | outputs/mbarara-concrete-factory-platform/src/db/postgres-schema.sql |
| Corrected browser download | http://127.0.0.1:5177/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf |

# Corrected Final Download

The browser-served file Mbarara_Integrated_Concrete_Factory_Master_Report.pdf is replaced with the corrected all-prompts merged PDF. It includes this supplement, the recent integrated module master report, and the earlier 194-page investor-grade report.

| Download item | Location |
| --- | --- |
| Direct browser download | http://127.0.0.1:5177/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf |
| Workspace output | outputs/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf |
| All-prompts merged copy | outputs/Mbarara_Integrated_Concrete_Factory_ALL_PROMPTS_MERGED_Report.pdf |

# Mbarara Integrated Concrete Products Factory

Investor, Engineering, Financial & Expansion Master Report 2026-2035

Prepared as a consolidated master report from the latest web platform modules: Financial Simulator, Inventory, Production Planning, Quality Control and Laboratory, ERP, Competitor and Market Intelligence, PostgreSQL structure, Product Costing, and Risk Register.

Currency: UGX only. Data confidence labels are preserved throughout. Market prices, contacts, competitor records, machinery costs, and supplier terms must be re-verified before investment decisions.

| | |
|---|---|
| Report file | Mbarara_Integrated_Concrete_Factory_Master_Report.pdf |
| Planning horizon | 2026-2035 |
| Location | Mbarara, Uganda |
| Data status | Estimated / Quotation Required / Needs verification unless explicitly marked Verified |

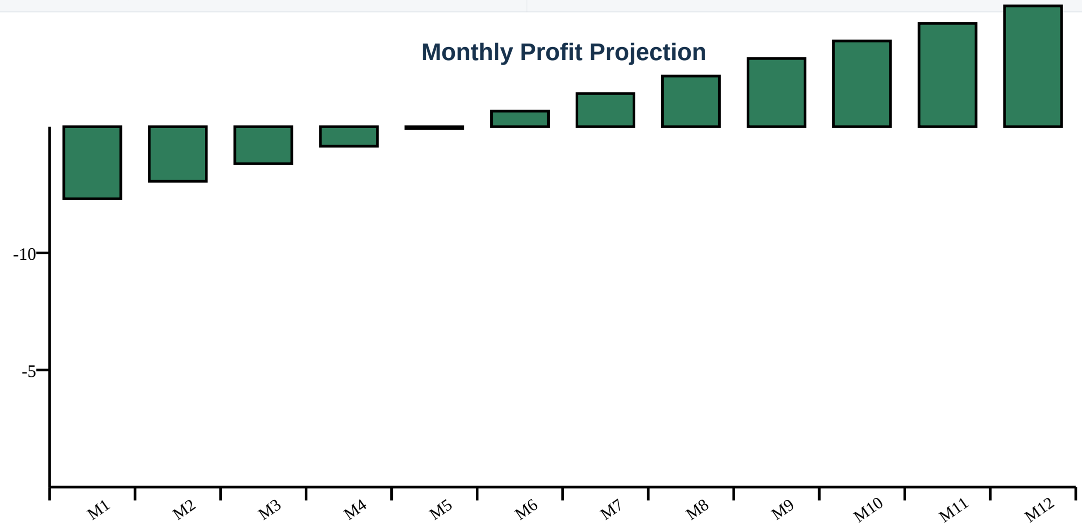
# Table of Contents

# 1. Executive Summary

The proposed Mbarara Integrated Concrete Products Factory is planned as an integrated concrete products factory serving Mbarara City and Western Uganda. The product set covers hollow blocks, solid blocks, pavers, kerbstones, drainage channels, culverts, and later ready-mix concrete.

The current integrated model shows monthly revenue of UGX 49,380,000, gross profit of UGX 16,460,553, and net profit of UGX -16,639,447. The UGX 20,000,000 monthly profit target is not yet reached under the seed assumptions.

| Metric | Value |
|---|---|
| Monthly revenue | UGX 49,380,000 |
| Monthly production/material cost | UGX 32,919,447 |
| Monthly expenses | UGX 33,100,000 |
| Gross profit | UGX 16,460,553 |
| Net profit | UGX -16,639,447 |
| Break-even revenue | UGX 99,296,662 |
| ROI estimate | -27.0% |
| UGX 20,000,000 target | Not reached |




# 2. Market Analysis

The first sales campaign should focus on areas with active housing, commercial construction, schools, roadworks, fuel stations, hotels, NGOs, churches, and hardware distribution. Customer demand records below are planning estimates and require field verification.

| |
|---|
| Target Western Uganda focus areas |
| Mbarara City, Rwampara, Isingiro, Ntungamo, Bushenyi, Ibanda, Kiruhura, Sheema, Lyantonde, Kabale, Rukungiri |

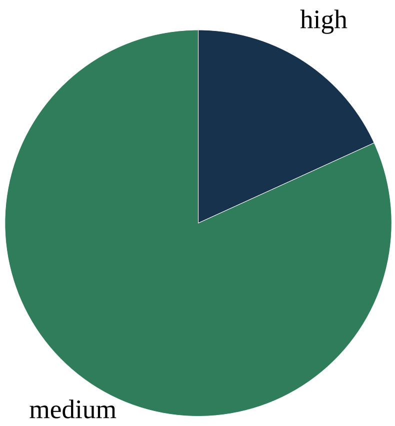
| Customer | Type | Location | Likely products | Demand | Reliability |
|---|---|---|---|---|---|
| Mbarara Heights Estate | property developers | Mbarara City | 6-inch blocks, 8-inch blocks, 60 mm pavers | 28,000 | medium |
| Nyamitanga Hardware | hardware stores | Mbarara City | 4-inch blocks, 6-inch blocks, 60 mm pavers | 14,000 | medium |
| Western Roads JV | government projects | Mbarara City and regional roads | kerbstones, drainage channels, culverts, 80 mm pavers | 5,000 | medium |
| Lake View Schools | schools | Ruti | 8-inch blocks, 60 mm pavers, drainage channels | 4,200 | high |
| Western Fuel Station Group | fuel stations | Mbarara, Ntungamo, Bushenyi | 80 mm pavers, kerbstones, drainage channels | 3,600 | medium |

# 3. Competitor Analysis

Competitor records are seeded for investigation only. No competitor record should be treated as verified until supported by website evidence, Facebook page review, phone call, quotation, field visit, marketplace check, or government record.

| Competitor | District | Business type | Products | Threat | Confidence |
|---|---|---|---|---|---|
| iConic Concrete | Mbarara City | precast supplier | hollow blocks, pavers, precast products | medium | Needs verification |
| Master Industries | Mbarara City | paver maker | pavers, blocks | medium | Needs verification |
| Stone Providers Ltd | Rwampara | quarry supplier | stone dust, aggregates | medium | Needs verification |
| Nicomah | Mbarara City | small block yard | hollow blocks | medium | Needs verification |
| BIMS Bloks Mbarara | Mbarara City | small block yard | hollow blocks, solid blocks | high | Needs verification |
| local block yards in Mbarara | Mbarara City | small block yard | 4-inch, 6-inch, 8-inch hollow blocks | high | Needs verification |
| local quarry suppliers around Western Uganda | Western Uganda | quarry supplier | stone dust, aggregates | medium | Needs verification |
| Hima Cement distributors | Mbarara City | cement distributor | cement | medium | Needs verification |
| Tororo Cement distributors | Mbarara City | cement distributor | cement | medium | Needs verification |
| Kampala Cement distributors | Mbarara City | cement distributor | cement | medium | Needs verification |
| Simba Cement distributors | Mbarara City | cement distributor | cement | medium | Needs verification |

**Threat Level Distribution**




# 4. Product Portfolio

| Product | Category | Unit | Daily volume | Planned price | Confidence |
|---|---|---|---|---|---|
| 4-inch hollow blocks | blocks | piece | 1,800 | UGX 1,800 | Estimated |
| 6-inch hollow blocks | blocks | piece | 2,200 | UGX 2,600 | Estimated |
| 8-inch hollow blocks | blocks | piece | 1,400 | UGX 3,400 | Estimated |
| solid blocks | blocks | piece | 950 | UGX 3,000 | Estimated |
| 60 mm pavers | pavers | piece | 3,200 | UGX 1,700 | Estimated |
| 80 mm pavers | pavers | piece | 2,300 | UGX 2,300 | Estimated |
| grass/permeable pavers | pavers | piece | 600 | UGX 4,200 | Quotation Required |
| cobblestone pavers | pavers | piece | 1,300 | UGX 2,100 | Estimated |
| kerbstones | precast | piece | 240 | UGX 18,000 | Estimated |
| drainage channels | precast | piece | 85 | UGX 55,000 | Quotation Required |

| Product | Category | Unit | Daily volume | Planned price | Confidence |
|---|---|---|---|---|---|
| culverts | precast | piece | 16 | UGX 180,000 | Quotation Required |
| ready-mix concrete | ready-mix | m3 | 35 | UGX 410,000 | Quotation Required |

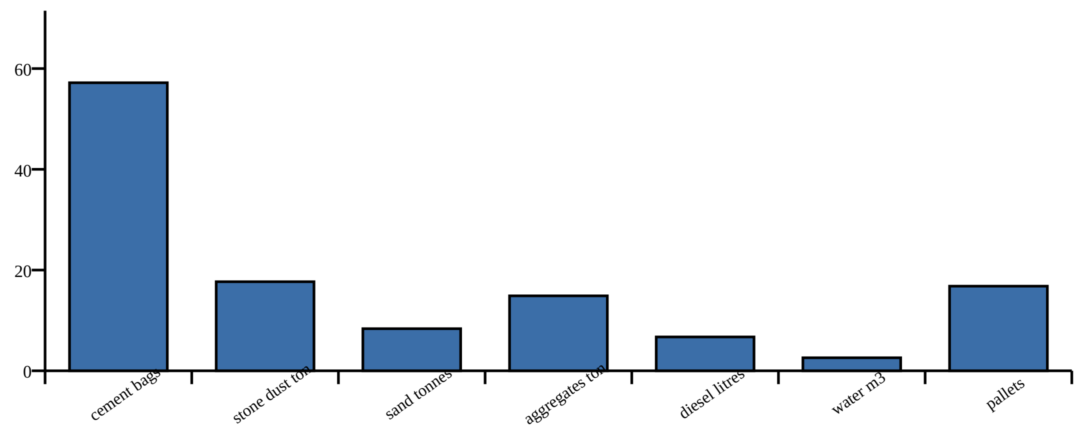
## 5. Product Mix Ratios

| Product | Mix ratio | Assumption label |
|---|---|---|
| 4-inch hollow blocks | 1 cement : 6 stone dust | Assumption: validate through batch trials, curing records, and compression tests. |
| 6-inch hollow blocks | 1 cement : 7 stone dust | Assumption: validate through batch trials, curing records, and compression tests. |
| 8-inch hollow blocks | 1 cement : 6 stone dust | Assumption: validate through batch trials, curing records, and compression tests. |
| solid blocks | 1 cement : 5 stone dust : 1 sand | Assumption: validate through batch trials, curing records, and compression tests. |
| 60 mm pavers | 1 cement : 3 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| 80 mm pavers | 1 cement : 2.5 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| grass/permeable pavers | 1 cement : 3 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| cobblestone pavers | 1 cement : 3 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| kerbstones | 1 cement : 2 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| drainage channels | 1 cement : 2 sand : 3 aggregate | Assumption: validate through batch trials, curing records, and compression tests. |
| culverts | 1 cement : 2 sand : 3 aggregate with reinforcement where specified | Assumption: validate through batch trials, curing records, and compression tests. |
| ready-mix concrete | Design mix by strength class | Assumption: validate through batch trials, curing records, and compression tests. |

## 6. Raw Material Consumption

Inventory is linked to production batches and financial costing. Reorder levels are planning values and should be adjusted after at least one month of measured usage.

| Item | Current qty | Unit | Stock value | Days remaining | Reorder level | Supplier |
|---|---|---|---|---|---|---|
| cement bags | 1,682.0 | bags | UGX 57,188,000 | 23.4 | 420.0 | Hima Cement distributors |
| stone dust tonnes | 393.0 | tonnes | UGX 17,685,000 | 21.8 | 90.0 | local quarry suppliers around Western Uganda |
| sand tonnes | 220.0 | tonnes | UGX 8,360,000 | 22.0 | 65.0 | local quarry suppliers around Western Uganda |
| aggregates tonnes | 248.0 | tonnes | UGX 14,880,000 | 20.7 | 80.0 | local quarry suppliers around Western Uganda |
| diesel litres | 1,200.0 | litres | UGX 6,720,000 | 21.8 | 360.0 | Mbarara diesel supplier |
| water m3 | 616.0 | m3 | UGX 2,587,200 | 25.7 | 160.0 | local quarry suppliers around Western Uganda |
| pallets | 480.0 | units | UGX 16,800,000 | 60.0 | 120.0 | local pallet supplier |
| moulds | 36.0 | units | UGX 27,000,000 | Non-consumable | 8.0 | machine/mould supplier |
| spare parts | 90.0 | units | UGX 11,250,000 | Non-consumable | 25.0 | machine spares supplier |




## 7. Machinery Analysis

| Machinery option | Suitable products | Role | Confidence |
|---|---|---|---|
| Semi-automatic block machine | Blocks and pavers | Phase-one line for repeatable output with manageable capex. | Quotation Required |
| Hydraulic paver press | 60 mm and 80 mm pavers | Improves finish, density, and institutional paving quality. | Quotation Required |
| Precast mould set | Kerbstones, channels, culverts | Enables higher-ticket road and drainage products. | Quotation Required |
| Batching plant and transit mixers | Ready-mix concrete | Phase-two expansion after verified demand and logistics capacity. | Quotation Required |

## 8. Factory Layout Summary

The recommended layout separates inbound materials, cement storage, batching/mixing, forming, curing yards, finished goods dispatch, laboratory, administration, spare parts store, and vehicle circulation. Final layout requires land survey, drainage design, and local authority approval.

| Zone | Function | Design note |
|---|---|---|
| Raw materials yard | Stone dust, sand, aggregates | Allow truck turning, drainage, and stockpile separation. |
| Covered cement store | Cement bags | Keep dry, secure, and close to batching. |
| Production bay | Mixing and forming | Minimize material handling and protect machines. |
| Curing yard | Curing and batch traceability | Track curing start date and batch ID. |
| QC laboratory | Tests and certificates | Keep records linked to production batches. |
| Dispatch zone | Customer loading | Separate from inbound quarry deliveries. |

## 9. Utility Consumption

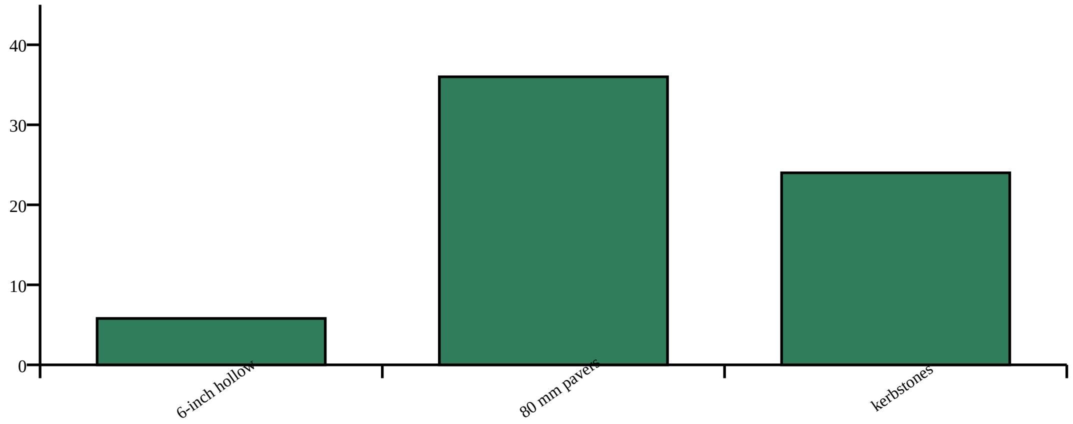
| Utility | Assumption | Cost driver | Confidence |
|---|---|---|---|
| Electricity | UGX 780 per kWh | Machine hours, mixer load, lighting | Estimated |
| Water | UGX 4,200 per m3 | Mixing, curing, cleaning | Estimated |
| Diesel | UGX 5,600 per litre | Loader, forklift, delivery, backup power | Estimated |

## 10. Laboratory and Quality Control

The QC module tracks each batch by batch ID, date, product, operator, material use, mix ratio, curing start date, curing age, compressive strength, water absorption, density, dimensions, visual defects, and approval status.

| Batch | Product | Strength | Absorption | Density | Dimensions | Defects | Approv |
|---|---|---|---|---|---|---|---|
| BATCH-2026-001 | 6-inch hollow blocks | 5.8 MPa | 8.2% | 1,720 kg/m3 | Pass | 2 | Passed |
| BATCH-2026-002 | 80 mm pavers | 36.0 MPa | 5.1% | 2,180 kg/m3 | Pass | 1 | Passed |
| BATCH-2026-003 | kerbstones | 24.0 MPa | 6.5% | 2,290 kg/m3 | Fail | 5 | Hold |

**Average Strength by Product (MPa)**




## 11. Legal and Regulatory Roadmap

| Office / process | Purpose | Timing |
|---|---|---|
| Mbarara City authority | Planning, building, trade, and local approvals | Pre-construction |
| NEMA | Environmental screening and mitigation | Pre-construction |
| Occupational safety office | Worker safety compliance and inspections | Before commissioning |
| URA | Tax registration, invoicing, and compliance | Before trading |
| UNBS / standards review | Product standards and certification pathway | Before making certification claims |
| Water and electricity utilities | Service connections and metering | Pre-commissioning |

## 12. Financial Model

| Product | Unit cost | Material | Utility | Labour | Price | Margin | Margin % | Confid |
|---|---|---|---|---|---|---|---|---|
| 4-inch hollow blocks | UGX 1,681 | UGX 1,493 | UGX 138 | UGX 50 | UGX 1,800 | UGX 119 | 6.6% | Estimat |
| 6-inch hollow blocks | UGX 1,681 | UGX 1,493 | UGX 138 | UGX 50 | UGX 2,600 | UGX 919 | 35.3% | Estimat |
| 8-inch hollow blocks | UGX 2,123 | UGX 1,935 | UGX 138 | UGX 50 | UGX 3,400 | UGX 1,277 | 37.6% | Estimat |
| solid blocks | UGX 2,400 | UGX 2,212 | UGX 138 | UGX 50 | UGX 3,000 | UGX 600 | 20.0% | Estimat |
| 60 mm pavers | UGX 1,995 | UGX 1,796 | UGX 148 | UGX 50 | UGX 1,700 | UGX -295 | -17.3% | Estimat |
| 80 mm pavers | UGX 2,267 | UGX 2,068 | UGX 148 | UGX 50 | UGX 2,300 | UGX 33 | 1.5% | Estimat |
| grass/permeable pavers | UGX 1,995 | UGX 1,796 | UGX 148 | UGX 50 | UGX 4,200 | UGX 2,205 | 52.5% | Quotati Require |
| cobblestone pavers | UGX 1,995 | UGX 1,796 | UGX 148 | UGX 50 | UGX 2,100 | UGX 105 | 5.0% | Estimat |
| kerbstones | UGX 8,849 | UGX 8,300 | UGX 499 | UGX 50 | UGX 18,000 | UGX 9,151 | 50.8% | Estimat |
| drainage channels | UGX 23,561 | UGX 22,000 | UGX 1,511 | UGX 50 | UGX 55,000 | UGX 31,439 | 57.2% | Quotati Require |

| Product | Unit cost | Material | Utility | Labour | Price | Margin | Margin % | Contro |
|---|---|---|---|---|---|---|---|---|
| culverts | UGX 80,890 | UGX 76,580 | UGX 4,260 | UGX 50 | UGX 180,000 | UGX 99,110 | 55.1% | Quotati Require |
| ready-mix concrete | UGX 351,922 | UGX 334,400 | UGX 17,472 | UGX 50 | UGX 410,000 | UGX 58,078 | 14.2% | Quotati Require |

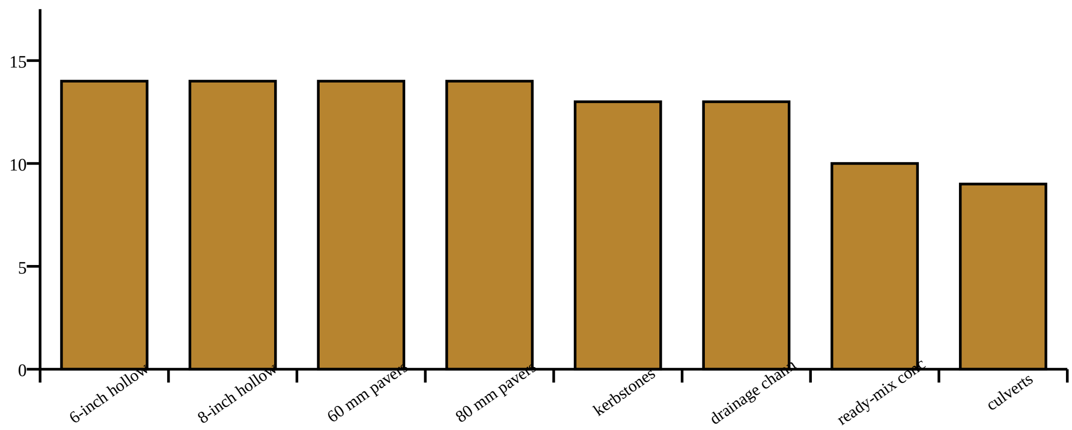
## 13. ERP Integration Summary

| Module | Connected records | Management output |
|---|---|---|
| Sales | Customers, products, unit price, delivery, paid amount | Daily sales, balances, gross revenue |
| Inventory | Items, suppliers, add stock, issue stock, reorder levels | Stock value, low-stock alerts, usage reports |
| Production | Batches, operators, material consumption, rejected products | Efficiency and batch traceability |
| Quality Control | Batch-linked laboratory tests | Pass/fail status, rejection rate, certificates |
| Expenses | Payroll, utilities, maintenance, transport | Monthly profit and supplier balances |

## 14. Market Gap Analysis

| Rank | Product | Demand | Competition | Margin | Logistics | Strategic | Opportunity |
|---|---|---|---|---|---|---|---|
| 1 | 6-inch hollow blocks | 9 | 7 | 6 | 3 | 9 | 14 |
| 2 | 8-inch hollow blocks | 8 | 6 | 7 | 3 | 8 | 14 |
| 3 | 60 mm pavers | 8 | 5 | 7 | 4 | 8 | 14 |
| 4 | 80 mm pavers | 7 | 5 | 8 | 4 | 8 | 14 |
| 5 | kerbstones | 6 | 4 | 8 | 5 | 8 | 13 |
| 6 | drainage channels | 6 | 3 | 8 | 6 | 8 | 13 |
| 7 | ready-mix concrete | 5 | 4 | 8 | 8 | 9 | 10 |
| 8 | culverts | 5 | 3 | 8 | 8 | 7 | 9 |
| 9 | grass/permeable pavers | 4 | 3 | 7 | 5 | 6 | 9 |
| 10 | cobblestone pavers | 5 | 4 | 6 | 4 | 6 | 9 |
| 11 | solid blocks | 5 | 6 | 5 | 3 | 5 | 6 |
| 12 | 4-inch hollow blocks | 6 | 7 | 4 | 3 | 5 | 5 |

**Opportunity Score by Product**




## 15. Risk Register

| Risk | Category | Likelihood | Impact | Mitigation |
|---|---|---|---|---|
| Market prices are unverified and may differ by district and delivery distance. | Market | High | High | Run weekly field price checks before final investment decisions. |
| Cement price volatility reduces margins. | Supply | Medium | High | Negotiate distributor credit terms and maintain supplier alternatives. |
| Insufficient curing and QC discipline leads to rejected blocks. | Quality | Medium | High | Enforce curing registers, batch traceability, and compression testing. |
| Equipment downtime disrupts monthly profit target. | Operations | Medium | High | Keep critical spares, preventive maintenance, and backup mixer capacity. |
| Delayed customer payments strain working capital. | Financial | Medium | Medium | Use customer credit limits and require deposits for delivery-heavy orders. |
| Permit or environmental compliance delay. | Regulatory | Low | High | Sequence NEMA, city authority, water, and occupational safety actions early. |

# 16. Implementation Roadmap 2026-2035

| Year | Roadmap focus |
|---|---|
| 2026 | Complete field verification, supplier quotations, permitting, pilot sales, and machine procurement. |
| 2027 | Commission blocks and pavers, formalize QC lab, and target Mbarara City and Rwampara. |
| 2028 | Add kerbstones and drainage channels, strengthen hardware distribution, and improve credit controls. |
| 2029 | Scale mould sets and secure framework supply agreements. |
| 2030 | Evaluate ready-mix feasibility and transit mixer economics. |
| 2031 | Expand to Ntungamo, Bushenyi, Ibanda, Kiruhura, Sheema, and Lyantonde. |
| 2032 | Build regional stockist network and contractor credit scoring. |
| 2033 | Add advanced precast products where municipal demand is verified. |
| 2034 | Prepare East African partnerships and logistics corridors. |
| 2035 | Evaluate second factory or satellite yard based on verified profitability. |

# 17. East African Expansion Strategy

Expansion should follow proof of quality, repeatable monthly profit, and working-capital discipline in Western Uganda. The recommended path is distribution-led before asset-heavy regional production expansion.

| Phase | Market logic | Action |
|---|---|---|
| Western Uganda consolidation | Shortest delivery routes and strongest relationships | Focus on Mbarara City, Rwampara, Isingiro, Ntungamo, Bushenyi, Ibanda, Kiruhura, Sheema, Lyantonde, Kabale, and Rukungiri. |
| Border market testing | Contractor and NGO demand may justify selective shipments | Use quotation-based sales before fixed depots. |
| Regional partnerships | Lower capex than immediate new factories | Partner with hardware stores and contractors first. |
| Second production base | Only after verified monthly profit and demand density | Trigger by multi-year contracts and reliable raw material supply. |

# 18. Appendices

## Data Confidence Labels

| Label | Meaning |
|---|---|
| Verified | Confirmed by reliable documentation, invoice, field check, or quotation. |
| Estimated | Planning value based on current model assumptions and prior project context. |
| Quotation Required | Do not use for procurement or investment until a supplier quotation is obtained. |
| Needs verification | Seed record requiring field visit, phone check, website check, or documentary proof. |

| Label | Meaning |
|---|---|
| Assumption | Model input that must be validated by quotations, lab tests, or site-specific engineering. |

## PostgreSQL Integration Tables

| Table | Purpose |
|---|---|
| products | Product portfolio, pricing, mix ratios, and active status. |
| suppliers | Cement, quarry, diesel, water, spares, and logistics suppliers. |
| customers | Demand, buying power, payment reliability, and target products. |
| production_batches | Material consumption, rejected products, curing, and machine hours. |
| inventory | Stock master data, unit costs, reorder levels, and daily consumption. |
| sales | Customer sales, unit price, delivery cost, paid amount, and balances. |
| expenses | Operating costs and supplier/payee balances. |
| quality_tests | Batch-linked strength, absorption, density, dimensions, defects, and approval. |
| competitor_records | Future market-intelligence extension with source tracking. |

## Investment Disclaimer

This master report is a planning document. It is not a substitute for engineering design, statutory approvals, audited financial statements, tax advice, legal advice, environmental approval, or binding supplier/customer quotations. Market prices, competitor details, contacts, GPS data, capacity estimates, and machinery costs must be re-verified before investment decisions.

INVESTOR-GRADE PRE-FEASIBILITY REPORT

# Mbarara Integrated Concrete Products Factory

A4 portrait consultancy report
Currency basis: UGX only
Commercial planning horizon: 2026-2035
Prepared: June 2026

Investor-grade pre-feasibility report generator v1.0

# Document Control

This PDF is generated from a reusable ReportLab generator and presents a pre-feasibility investment case for a concrete products factory serving Mbarara, western Uganda and selected East African corridors.

All monetary values are stated in UGX only. Values are indicative planning assumptions, not quotations. Final investment decisions should be supported by supplier quotations, professional engineering, tax advice, legal review, geotechnical work and customer due diligence.

| Field | Record |
|---|---|
| Project | Mbarara Integrated Concrete Products Factory |
| Location basis | Mbarara City / western Uganda industrial catchment |
| Report form | A4 portrait investor-grade PDF report |
| Planning horizon | 2026-2035 |
| Financial basis | UGX nominal, before financing structure unless noted |
| Prepared | June 2026 |

**Investor use: screening, bankability discussion, board preparation, lender questions and procurement planning.**

# Important Notice and Limitations

This report is intentionally investor-ready in structure but remains a pre-feasibility model. It uses public-source market context and internally generated operating assumptions for product mix, throughput, CAPEX, OPEX and 10-year forecast.

The model should be refreshed before financial close with binding supplier quotes, local construction BoQs, electricity connection studies, water supply confirmation, tax advice, product certification requirements and signed customer off-take indications.

| Limitation | Implication | Recommended diligence action |
|---|---|---|
| Market sizing | Demand is inferred from construction, urbanization and infrastructure drivers. | Run contractor interviews and collect public tender pipeline data. |
| CAPEX | Indicative equipment and civil costs can move materially. | Obtain three supplier quotations and a local quantity surveyor estimate. |
| Regulatory | Approval route may vary with site and environmental sensitivity. | Screen site with Mbarara City, NEMA and relevant lead agencies. |
| Tax | Model includes generic tax assumptions. | Confirm incentives, VAT treatment and depreciation allowances with URA/UIA advisors. |

# Table of Contents

Chapter and major-section page numbers are generated automatically by the PDF builder.

# 1. Executive Summary

**Recommended base case: a staged integrated factory with automatic block/paver production, dedicated kerb/channel/culvert capability and a ready-mix plant introduced after market traction is proven.**

| FUNDING ENVELOPE | OPERATING BREAK-EVEN | TARGET MARKETS |
|---|---|---|
| **UGX 36.13 bn** | **Year 3** | **Western Uganda** |
| includes contingency and opening working capital | base case reaches positive operating leverage after ramp-up | Mbarara, Bushenyi, Ntungamo, Kabale, Isingiro and cross-border corridors |

The investment thesis is to establish a quality-led, mechanized concrete products factory in Mbarara that can serve residential, institutional, road, drainage and commercial concrete demand with reliable supply, certified quality and disciplined dispatch.

**1 Investor lens**

What the chapter means for return, risk and bankability.

**2 Operating lens**

Factory routines, people, quality and throughput implications.

**3 Decision lens**

Actions required before financial close and procurement.

# Executive Dashboard

The factory is positioned as a formal, certified and reliable alternative to fragmented manual production. The model rewards scale, disciplined curing, reliable delivery and a portfolio that balances daily building materials with infrastructure-oriented products.

The base case assumes a 2026 setup year, commercial launch in 2027 and progressive ramp to regional distribution by 2035.

**TOTAL FUNDING**

**UGX 36.13 bn**

CAPEX, contingency and initial working capital.

**MODEL IRR**

**8.2%**

Indicative unlevered project IRR from 2026-2035 cash flows.

**NPV AT 18%**

**UGX -14.45 bn**

Discounted unlevered cash flow on base-case assumptions.

**YEAR 2035 REVENUE**

**UGX 58.60 bn**

Nominal revenue across blocks, pavers, civils and ready-mix.

**EBITDA MARGIN 2035**

**28.4%**

Mature margin depends on quality yield and utilization.

**PRODUCT FAMILIES**

**13**

blocks, pavers, kerbs, channels, culverts and RMC.

# Investment Rationale

Mbarara is a strategic production base because it combines western Uganda demand, transport access to construction corridors and proximity to Rwanda, DRC-facing routes and fast-growing secondary towns.

The investor case is not simply to make blocks. The value proposition is a formal building materials platform: documented mix designs, cube testing, uniform dimensions, reliable curing, scheduled deliveries, branded stock availability and contractor credit discipline.

The recommended approach is staged. Commission the core block and paver line first, stabilize quality and cash conversion, then add ready-mix and heavier civil products as anchor contracts and dispatch capability mature.

**1 Scale**

Mechanized output lowers unit cost versus manual yards once utilization rises.

**2 Quality**

Lab-backed batches help win institutional and contractor confidence.

**3 Portfolio**

Blocks drive frequency; pavers and civils raise margin depth.

**4 Location**

Mbarara supports western Uganda and cross-border expansion options.

## Recommended Strategic Configuration

The base case uses a fully automatic vibropress production line with modular molds, covered curing, a basic but credible concrete laboratory, yard handling equipment and a sales/distribution model anchored by contractor relationships.

Ready-mix concrete should be planned into the site from day one but commercially scaled once the factory has proven dispatch discipline, quality records, water control and fleet management.

| Decision area | Recommendation | Reason |
|---|---|---|
| Production line | Automatic vibropress core line | Improves dimensional consistency, production speed and traceability. |
| Curing | Covered wet curing plus controlled chambers | Protects strength development and minimizes rejected batches. |
| RMC | Install in phase 2 or late phase 1 | Higher working-capital and fleet demands require disciplined dispatch. |
| Sales model | Contractor, municipal and dealer channels | Balances volume orders with repeat trade and route coverage. |
| Quality | UNBS/NBRB-oriented QC system | Differentiates against informal supply and supports institutional tenders. |

# Top Ten Investor Questions

The questions below should be answered before financial close. They convert the report from pre-feasibility into a lender-ready investment memorandum.

| No. | Question | Evidence required |
|---|---|---|
| 1 | What site will be secured? | Title/lease, zoning confirmation, access, drainage, geotechnical and utilities route. |
| 2 | Which equipment supplier is bankable? | Technical offer, warranty, local support, mold list, spares and commissioning plan. |
| 3 | Who are anchor customers? | Letters of intent from contractors, estates, road contractors and institutional buyers. |
| 4 | How will quality be certified? | UNBS product route, lab plan, cube testing protocol and retained sample register. |
| 5 | What is the working capital cycle? | Credit policy, receivable aging assumptions, stock turns and customer payment terms. |
| 6 | How resilient is power and water? | UEDCL connection assessment, standby generation, NWSC/borehole plan and recycling. |
| 7 | Can price survive informal competition? | Delivered-cost analysis, contractor interviews and certification premium testing. |
| 8 | What approvals are gating? | NEMA screening, city permits, URSB/URA/NSSF, OHS and trade license calendar. |
| 9 | How will management control shrinkage? | Weighbridge, ERP, batch tickets, dispatch reconciliations and stock takes. |
| 10 | When should expansion start? | Only after service level, receivable days and quality pass rate remain within thresholds. |

# 2. Market Analysis

**Public data from the World Bank and UBOS points to a large, young and growing economy with industry, manufacturing and construction as important growth channels. The factory should treat the market as a corridor system: urban Mbarara plus western districts, road projects, institutions and selected export-adjacent routes.**

UGANDA POPULATION

**45.9m**

UBOS 2024 Census portal total population

GROWTH CONTEXT

**6%+ GDP**

World Bank reported strong growth into 2025/2026

MARKET FORM

**Fragmented**

formal producers, local yards, dealers and site casting compete

Uganda's economy and construction demand provide a supportive context for local building-material production, while Mbarara gives the project a regional distribution base rather than a single-city market.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Demand Drivers

The most important demand drivers are urban housing, estate development, schools and clinics, road drainage, parking and paving, commercial slabs, small industrial sheds, and public-sector works.

Concrete products are heavy and low value per kilogram. This gives a local factory a delivered-cost advantage if it can maintain quality and avoid unreliable lead times.

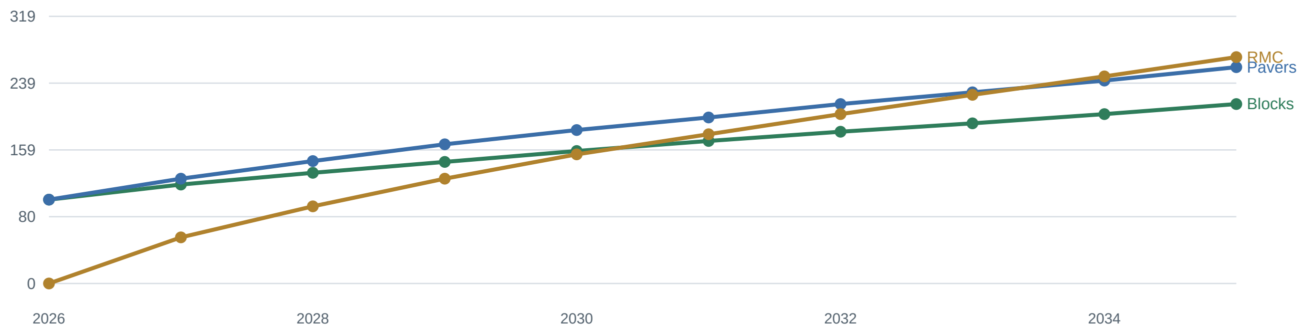
| Demand driver | Products pulled | Commercial implication |
|---|---|---|
| Urban housing | 4-inch, 6-inch and 8-inch blocks | High-frequency demand; dealer and contractor relationships matter. |
| Commercial estates | blocks, pavers, ready-mix | Requires credit control, delivery planning and consistent dimensions. |
| Municipal roads | kerbs, pavers, drainage channels | Needs tender documents, quality files and delivery scheduling. |
| Rural access roads | culverts, channels, concrete | Batch traceability and reinforcement compliance are key. |
| Tourism and hospitality | cobblestone, hexagonal and permeable pavers | Premium visual products can raise margin. |

# Macro Context

The World Bank describes Uganda as a young, growing country and reported broad-based economic performance with industry, manufacturing and construction among strong areas. This does not automatically translate into factory sales, but it supports a positive screening view.

UBOS maintains annual GDP, quarterly GDP and construction input price publications. These should become the monthly reference set for management once the factory is operational.

**Indicative demand index by segment, 2026-2035 (2026 = 100)**



# Western Uganda Catchment

Mbarara should be treated as the operating hub for a wider catchment rather than the only source of demand. The immediate ring includes Mbarara City, Isingiro, Kiruhura, Ibanda and Bushenyi; the second ring extends to Ntungamo, Kabale, Rukungiri, Kasese and Fort Portal depending on transport economics.

The factory should price by delivered zone, not one ex-works price. Blocks and kerbs are freight-sensitive; RMC is time-sensitive; pavers can travel farther if packed well and loaded efficiently.

| Zone | Target radius | Best-fit products | Route logic |
|---|---|---|---|
| Core Mbarara | 0-25 km | blocks, pavers, RMC | same-day dispatch and repeat contractors |
| Western ring | 25-100 km | blocks, kerbs, pavers | scheduled dealer drops and project deliveries |
| Highland route | 100-180 km | pavers, culverts | higher-value loads and less frequent drops |
| Cross-border prep | route-based | pavers, culverts | depot or partner model after domestic proof |

# Customer Segmentation

The commercial plan should separate cash buyers, contractors, institutions, public projects, dealers and ready-mix customers. Each segment has different margin, credit and service requirements.

Management should avoid giving contractor credit before the plant has a live receivables dashboard and signed delivery acceptance protocol.

| Segment | Buying behavior | Service requirement | Credit stance |
|---|---|---|---|
| Small builders | frequent small lots | visible stock and loading speed | cash or mobile payment |
| Contractors | bulk orders by project phase | delivery accuracy and documents | approved limits only |
| Institutions | planned procurement | quality files and tax compliance | formal payment cycle |
| Dealers | replenishment loads | consistent margins and packaging | secured rotation |
| RMC customers | day-specific pours | slump control and delivery tickets | deposit and pour sign-off |

# Pricing Architecture

The factory should publish ex-works list prices, delivered zone prices and project quotation rules. Discounting should be tied to volume, payment speed and pallet return discipline.

A basic price discipline rule is that any discount below standard gross margin must be approved by finance with a reason code.

| Price layer | Purpose | Control |
|---|---|---|
| Ex-works list | Transparent baseline for dealers and contractors | Reviewed monthly against cement and fuel. |
| Delivered zone | Captures freight and loading costs | Distance bands with minimum load sizes. |
| Project quote | Supports tenders and large jobs | Validity period and escalation clause. |
| Cash discount | Improves liquidity | Only for prompt payment or advance deposits. |

# Procurement Pipeline Targets

The first 18 months should prioritize conversion of a practical pipeline rather than generic brand awareness. The best early customers are those that value reliable dimensions, delivery and documentation.

The sales team should maintain a rolling pipeline by project stage: identified, qualified, quoted, negotiated, won, delivered and collected.

| Pipeline source | Products | Evidence to collect |
|---|---|---|
| Residential estates | blocks, pavers | project size, build stages, creditworthiness |
| Schools and clinics | blocks, channels, pavers | procurement calendar and technical requirements |
| Road contractors | kerbs, channels, culverts | BOQ, specification and delivery windows |
| Commercial slabs | RMC | pour schedule, pump needs and site access |

# Market Data to Refresh Monthly

Investor-grade operation requires a market dashboard. The factory should not rely on annual reports alone; it should track leading indicators that move product demand and margin.

The initial dashboard should include cement landed price, fuel price, aggregate cost, construction input index, tender pipeline, receivable days and competitive delivered prices.

| Indicator | Source or owner | Decision supported |
|---|---|---|
| Cement landed price | procurement | pricing and gross margin |
| Fuel and transport | logistics | delivered-zone pricing |
| Construction input index | UBOS publications | escalation clauses and budget review |
| Tender pipeline | sales manager | mold planning and inventory |
| Receivable days | finance | credit limits and cash conversion |

# Market Visual Placeholder

The final investment memorandum should insert a GIS-style map showing Mbarara, production site, delivery rings, industrial parks, quarry sources, cement supply routes and target cross-border routes.

**MAP PLACEHOLDER**

Recommended insert: catchment and logistics map with 25 km, 100 km and 180 km delivery rings.

**Placeholder retained intentionally for investor-ready layout and future diligence evidence.**

# 3. Competitor Analysis

**The project should not compete only on headline unit price. Its defensible angle is certified quality, dependable stock, documented testing, delivery scheduling, product breadth and contractor service.**

COMPETITOR LAYERS

**5**

informal, regional, national, cement-linked and site-cast

PRIMARY THREAT

**price**

informal yards can undercut if customers ignore quality

PRIMARY EDGE

**reliability**

consistent product, logistics and QC records

Competition is best understood as a layered market: informal local block yards, regional concrete product suppliers, national automated producers, cement-linked building-material brands and site-cast alternatives.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Competitor Landscape

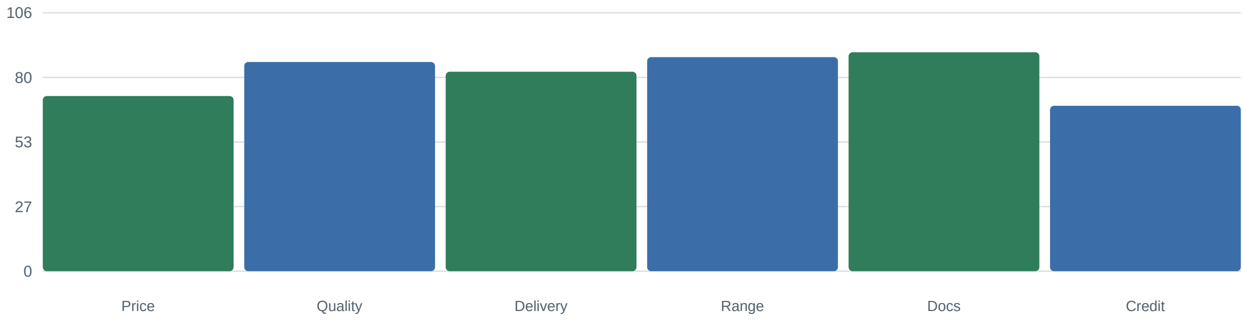
Public web references identify western Uganda suppliers such as Nicomah and national players such as Master Industries, plus cement producers with product breadth such as Hima and Tororo. This landscape confirms that buyers already understand concrete products, but quality and delivery consistency remain opportunities for differentiation.

| Layer | Examples | Strength | Weakness to exploit |
|---|---|---|---|
| Informal local yards | manual block makers and small paver yards | low visible price and proximity | variable curing, dimensions and documentation |
| Western suppliers | Nicomah-type material suppliers | local routes and customer familiarity | may lack integrated plant depth |
| National automated producers | Master Industries-type suppliers | scale, product breadth and brand | freight disadvantage into western districts |
| Cement-linked brands | Hima/Tororo product ecosystems | cement credibility and technical know-how | not always local stock across all products |
| Site casting | contractor-made kerbs, channels and slabs | custom fit and perceived control | quality inconsistency and supervision burden |

# Competitive Positioning Map

The recommended market position is upper-middle: not the cheapest producer, but reliable enough for contractors and institutions to justify a quality premium and lower failure risk.

**Illustrative competitor score by purchase criterion**



# Differentiation Levers

The factory should convert quality into commercial language: fewer breakages, fewer site delays, predictable dimensions, support for inspections and cleaner documentation for institutions.

| Lever | Operational requirement | Buyer benefit |
|---|---|---|
| Stamped batch lot | lot number, date and mix family | traceability for disputes and quality assurance |
| Published curing standard | minimum curing days by product | reduced breakage and better confidence |
| Delivery booking window | dispatch board and fleet control | contractors plan labor with less idle time |
| Contractor technical sheet | product specs, use cases, installation notes | reduces misuse and callbacks |
| Credit scorecard | finance-approved terms | protects cash while supporting repeat customers |

# Competitor Response Scenarios

Competitor response should be expected after launch, especially if the factory gains institutional contracts. The response plan should be prepared before discounts start.

| Scenario | Likely trigger | Response |
|---|---|---|
| Price undercut | large block order | hold list price; offer delivery reliability and quality documentation, not blind discounting |
| Claims on strength | contractor dispute | produce cube results, retained samples and batch ticket record |
| Dealer lock-in | route expansion | target unserved dealers and direct contractors before exclusive dealer deals |
| Fast imitation | popular paver profile | rotate colors, introduce installation support and maintain mold condition |
| Credit pressure | contractor demands terms | tie credit to signed acceptance, aging performance and project owner reputation |

# Competitor Due Diligence Checklist

Before final approval, the sponsor should complete a field competitor survey. The goal is to identify delivered prices, payment terms, breakage rates, stockouts and product profiles actually available in western Uganda.

| Check | Question | Evidence |
|---|---|---|
| Price | What is actual delivered price by zone? | quote screenshots, receipts and contractor interviews |
| Quality | What products break or vary most? | site photos and mason feedback |
| Lead time | How often do buyers wait? | contractor procurement logs |
| Credit | Who offers terms and to whom? | dealer interviews and payment practice |
| Range | Which profiles are hard to source? | dealer shelves and project BOQs |
| Standards | Who can show test data? | product certificates and lab records |

# Competitor Profile: Informal block makers

Informal block makers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

| Dimension | Assessment |
|---|---|
| Customer pull | high for routine products, moderate for premium profiles |
| Likely price behavior | discounting around large tenders and seasonal slowdowns |
| Operational watchpoint | stock availability, cure time, product damage and delivery reliability |
| Recommended response | quality files, reliable loading, zone-based delivery and disciplined credit |

# Competitor Profile: Western material suppliers

Western material suppliers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

| Dimension | Assessment |
|---|---|
| Customer pull | high for routine products, moderate for premium profiles |
| Likely price behavior | discounting around large tenders and seasonal slowdowns |
| Operational watchpoint | stock availability, cure time, product damage and delivery reliability |
| Recommended response | quality files, reliable loading, zone-based delivery and disciplined credit |

# Competitor Profile: National automated producers

National automated producers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

| Dimension | Assessment |
|---|---|
| Customer pull | high for routine products, moderate for premium profiles |
| Likely price behavior | discounting around large tenders and seasonal slowdowns |
| Operational watchpoint | stock availability, cure time, product damage and delivery reliability |
| Recommended response | quality files, reliable loading, zone-based delivery and disciplined credit |

# Competitor Profile: Cement-linked producers

Cement-linked producers should be tracked as a separate competitor group because buyer behavior, price flexibility and service promise differ materially from the other groups.

The factory's response should be evidence-led: compare total installed cost, breakage, delivery risk and documentation, not just ex-yard unit price.

| Dimension | Assessment |
|---|---|
| Customer pull | high for routine products, moderate for premium profiles |
| Likely price behavior | discounting around large tenders and seasonal slowdowns |
| Operational watchpoint | stock availability, cure time, product damage and delivery reliability |
| Recommended response | quality files, reliable loading, zone-based delivery and disciplined credit |

# 4. Product Portfolio

**The initial production plan should prioritize blocks and standard pavers for utilization, while maintaining molds and process capability for kerbs, channels, culverts and ready-mix as contract opportunities emerge.**

PRODUCT FAMILIES

**13**

all requested products included

CORE VOLUME

**Blocks**

daily builder demand and dealer replenishment

MARGIN DEPTH

**Pavers/civils**

premium profiles and infrastructure applications

The product portfolio combines high-frequency masonry units with higher-margin paving and infrastructure products. This gives the factory multiple demand cycles and reduces reliance on one buyer segment.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Portfolio Master Table

Indicative prices are model inputs for screening only and should be refreshed using Mbarara delivered quotations before financing. Pavers are priced by square metre; ready-mix by cubic metre; culverts and kerbs by unit.

| Product | Code | Format | Primary buyers | Indicative unit price |
|---|---|---|---|---|
| 4-inch hollow blocks | B100 | 100 x 200 x 400 mm | internal partitions, low-load walls, fast estate infill | UGX 1,900 |
| 6-inch hollow blocks | B150 | 150 x 200 x 400 mm | external walls, boundary walls, schools, clinics | UGX 2,600 |
| 8-inch hollow blocks | B200 | 200 x 200 x 400 mm | retaining, commercial walls, institutional projects | UGX 3,400 |
| solid concrete blocks | BSOL | 100/150/200 mm families | load-bearing details, plinths, hard-wearing walls | UGX 3,200 |
| zig-zag pavers | PZZ | 60/80 mm thickness | driveways, parking, walkways and estates | UGX 43,000 |
| double-T pavers | PDT | 60/80 mm thickness | heavy interlocking yards, schools and stations | UGX 45,500 |
| hexagonal pavers | PHX | 60 mm standard | landscape, hotels, pedestrian precincts | UGX 47,000 |
| cobblestone pavers | PCB | 60/80 mm aesthetic setts | premium courtyards, resorts, walkways | UGX 52,000 |
| grass/permeable pavers | PGP | 80/100 mm permeable grid | green parking, stormwater-sensitive sites | UGX 56,000 |
| kerbstones | KRB | 1000 mm road profiles | roads, estates, car parks and drainage edges | UGX 23,500 |
| drainage channels | DCH | 600/1000 mm channel units | roads, industrial drainage, estates | UGX 78,000 |
| culverts | CLV | 300-1200 mm diameter | rural roads, city drainage, farm crossings | UGX 275,000 |
| ready-mix concrete | RMC | M15-M40 supplied by m3 | commercial slabs, foundations, infrastructure | UGX 365,000 |

# Portfolio Role Matrix

The role of each product is different. A healthy factory should not optimize only for output volume; it should balance utilization, cash conversion, margin and customer acquisition.

| Portfolio role | Products | Commercial purpose | Management metric |
|---|---|---|---|
| Traffic builder | 4-inch and 6-inch hollow blocks | frequent sales and dealer pull | daily units sold and breakage |
| Specification anchor | 8-inch blocks, solid blocks | institutional and structural applications | cube pass rate and dimensions |
| Margin enhancer | double-T, hexagonal, cobblestone pavers | premium landscape and parking work | gross margin by profile |
| Infrastructure ticket | kerbs, channels, culverts | road and drainage contracts | tender win rate and delivery completion |
| Service platform | ready-mix concrete | commercial pours and large sites | on-time pours and rejected loads |

# Product Sheet: 4-inch hollow blocks

4-inch hollow blocks (B100) is modeled as a formal factory product with standard format of 100 x 200 x 400 mm. The primary market is internal partitions, low-load walls, fast estate infill.

Indicative strength basis: 3.5-5.0 MPa target subject to UNBS testing. Indicative mix starting point: 1 cement : 6-8 graded sand/stone dust. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | internal partitions, low-load walls, fast estate infill |
| Indicative unit price | UGX 1,900 |
| Indicative daily capacity | 5,200 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for 4-inch hollow blocks.

# Product Sheet: 6-inch hollow blocks

6-inch hollow blocks (B150) is modeled as a formal factory product with standard format of 150 x 200 x 400 mm. The primary market is external walls, boundary walls, schools, clinics.

Indicative strength basis: 5.0-7.5 MPa target. Indicative mix starting point: 1 cement : 5.5-7 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
| --- | --- |
| Target customers | external walls, boundary walls, schools, clinics |
| Indicative unit price | UGX 2,600 |
| Indicative daily capacity | 4,300 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for 6-inch hollow blocks.

# Product Sheet: 8-inch hollow blocks

8-inch hollow blocks (B200) is modeled as a formal factory product with standard format of 200 x 200 x 400 mm. The primary market is retaining, commercial walls, institutional projects.

Indicative strength basis: 7.5-10.0 MPa target. Indicative mix starting point: 1 cement : 5-6.5 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | retaining, commercial walls, institutional projects |
| Indicative unit price | UGX 3,400 |
| Indicative daily capacity | 3,300 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for 8-inch hollow blocks.

# Product Sheet: solid concrete blocks

solid concrete blocks (BSOL) is modeled as a formal factory product with standard format of 100/150/200 mm families. The primary market is load-bearing details, plinths, hard-wearing walls.

Indicative strength basis: 7.5-12.5 MPa target. Indicative mix starting point: 1 cement : 4.5-6 aggregate blend. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | load-bearing details, plinths, hard-wearing walls |
| Indicative unit price | UGX 3,200 |
| Indicative daily capacity | 2,600 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |

**PRODUCT IMAGE PLACEHOLDER**

Insert product photo/rendering for solid concrete blocks.

# Product Sheet: zig-zag pavers

zig-zag pavers (PZZ) is modeled as a formal factory product with standard format of 60/80 mm thickness. The primary market is driveways, parking, walkways and estates.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 2 sand : 3 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | driveways, parking, walkways and estates |
| Indicative unit price | UGX 43,000 |
| Indicative daily capacity | 780 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for zig-zag pavers.

# Product Sheet: double-T pavers

double-T pavers (PDT) is modeled as a formal factory product with standard format of 60/80 mm thickness. The primary market is heavy interlocking yards, schools and stations.

Indicative strength basis: 40-50 MPa target. Indicative mix starting point: 1 cement : 1.8 sand : 2.7 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
| --- | --- |
| Target customers | heavy interlocking yards, schools and stations |
| Indicative unit price | UGX 45,500 |
| Indicative daily capacity | 690 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for double-T pavers.

# Product Sheet: hexagonal pavers

hexagonal pavers (PHX) is modeled as a formal factory product with standard format of 60 mm standard. The primary market is landscape, hotels, pedestrian precincts.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 2 sand : 3 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | landscape, hotels, pedestrian precincts |
| Indicative unit price | UGX 47,000 |
| Indicative daily capacity | 510 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |

**PRODUCT IMAGE PLACEHOLDER**

Insert product photo/rendering for hexagonal pavers.

# Product Sheet: cobblestone pavers

cobblestone pavers (PCB) is modeled as a formal factory product with standard format of 60/80 mm aesthetic setts. The primary market is premium courtyards, resorts, walkways.

Indicative strength basis: 35-45 MPa target. Indicative mix starting point: 1 cement : 1.8 sand : 2.8 aggregate. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | premium courtyards, resorts, walkways |
| Indicative unit price | UGX 52,000 |
| Indicative daily capacity | 390 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for cobblestone pavers.

# Product Sheet: grass/permeable pavers

grass/permeable pavers (PGP) is modeled as a formal factory product with standard format of 80/100 mm permeable grid. The primary market is green parking, stormwater-sensitive sites.

Indicative strength basis: 30-40 MPa target with void control. Indicative mix starting point: gap-graded aggregate with controlled paste. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | green parking, stormwater-sensitive sites |
| Indicative unit price | UGX 56,000 |
| Indicative daily capacity | 300 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for grass/permeable pavers.

# Product Sheet: kerbstones

kerbstones (KRB) is modeled as a formal factory product with standard format of 1000 mm road profiles. The primary market is roads, estates, car parks and drainage edges.

Indicative strength basis: M25-M35 concrete target. Indicative mix starting point: M25-M35 design mix with reinforcement where required. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | roads, estates, car parks and drainage edges |
| Indicative unit price | UGX 23,500 |
| Indicative daily capacity | 560 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for kerbstones.

# Product Sheet: drainage channels

drainage channels (DCH) is modeled as a formal factory product with standard format of 600/1000 mm channel units. The primary market is roads, industrial drainage, estates.

Indicative strength basis: M25-M35 reinforced concrete. Indicative mix starting point: M25-M35 design mix; steel cages as specified. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
| --- | --- |
| Target customers | roads, industrial drainage, estates |
| Indicative unit price | UGX 78,000 |
| Indicative daily capacity | 95 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for drainage channels.

# Product Sheet: culverts

culverts (CLV) is modeled as a formal factory product with standard format of 300-1200 mm diameter. The primary market is rural roads, city drainage, farm crossings.

Indicative strength basis: reinforced concrete pipe class by design. Indicative mix starting point: M30+ with cage reinforcement and steam/wet curing option. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | rural roads, city drainage, farm crossings |
| Indicative unit price | UGX 275,000 |
| Indicative daily capacity | 25 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |

**PRODUCT IMAGE PLACEHOLDER**

Insert product photo/rendering for culverts.

# Product Sheet: ready-mix concrete

ready-mix concrete (RMC) is modeled as a formal factory product with standard format of M15-M40 supplied by m3. The primary market is commercial slabs, foundations, infrastructure.

Indicative strength basis: designed by grade with cube verification. Indicative mix starting point: approved design mix by grade and slump. Final mix designs must be validated by laboratory trials, cube tests, curing performance and applicable standards.

| Attribute | Investor-grade operating note |
|---|---|
| Target customers | commercial slabs, foundations, infrastructure |
| Indicative unit price | UGX 365,000 |
| Indicative daily capacity | 95 units or m2/m3 basis as applicable |
| Quality control | dimension checks, batch records, visual defects, curing status and compressive strength testing |
| Sales note | quote ex-works and delivered-zone prices separately; enforce payment terms by customer class |


Insert product photo/rendering for ready-mix concrete.

# Portfolio Launch Sequence

The launch sequence should favor products that stabilize the core production line before expanding into highly scheduled or engineered products.

**Recommended product sequencing**

Blocks → Standard pavers → Kerbs → Channels → Ready-mix → Culverts → Premium permeable range

**Stage-gate rule: do not launch the next family until batch records, curing compliance and receivable days remain within management thresholds for two consecutive months.**

# 5. Mix Ratios

**The NBRB concrete guidance referenced in this report emphasizes controlled water-cement ratio, cube sampling, curing and mechanical compaction. These principles are built into the proposed factory quality plan.**

WATER-CEMENT CONTROL

**<= 0.50**

NBRB guidance for dry aggregates

CUBE SAMPLING

**6 per mix**

3 tested at 7 days and 3 at 28 days

CURING TARGET

**14 days**

minimum curing reference from NBRB guidance

Mix ratios in this report are starting points for feasibility and mold trials, not final engineering specifications. Final ratios must be approved through controlled batch testing and strength results.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Mix Ratio Summary

The factory should move from volume ratios to weigh-batched recipes once production stabilizes. Moisture correction is critical because sand and stone dust moisture directly affects strength, texture and demolding.

| Product | Indicative mix by volume | Water control | QC trigger |
|---|---|---|---|
| 4-inch hollow blocks | 1 cement : 6-8 graded sand/stone dust | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| 6-inch hollow blocks | 1 cement : 5.5-7 aggregate blend | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| 8-inch hollow blocks | 1 cement : 5-6.5 aggregate blend | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| solid concrete blocks | 1 cement : 4.5-6 aggregate blend | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| zig-zag pavers | 1 cement : 2 sand : 3 aggregate | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| double-T pavers | 1 cement : 1.8 sand : 2.7 aggregate | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| hexagonal pavers | 1 cement : 2 sand : 3 aggregate | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| cobblestone pavers | 1 cement : 1.8 sand : 2.8 aggregate | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| grass/permeable pavers | gap-graded aggregate with controlled paste | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| kerbstones | M25-M35 design mix with reinforcement where required | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| drainage channels | M25-M35 design mix; steel cages as specified | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| culverts | M30+ with cage reinforcement and steam/wet curing option | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |
| ready-mix concrete | approved design mix by grade and slump | target w/c <= 0.50 where applicable | trial cubes and dimensional pass |

# Mix Design Trial Sheet: Masonry Blocks

Products covered: 4-inch hollow blocks, 6-inch hollow blocks, 8-inch hollow blocks and solid concrete blocks. Starting mix basis: 1 cement : 4.5-8 aggregate blend depending on strength target and unit thickness. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

| Trial field | Entry |
|---|---|
| Cement type | CEM class to be recorded by supplier batch |
| Aggregate blend | sand, stone dust and coarse aggregate proportions by weight |
| Water addition | metered water plus moisture adjustment |
| Admixture | only after baseline mix is proven |
| Cube/testing plan | minimum 7-day and 28-day checks; retain batch samples |
| QC focus | prioritize dimensional tolerance, demolding quality, breakage rate and 7/28-day cube trend |
| Production sign-off | plant manager, QC lead and maintenance lead |

# Mix Design Trial Sheet: Standard Pavers

Products covered: zig-zag, double-T and hexagonal pavers. Starting mix basis: 1 cement : 1.8-2 sand : 2.7-3 aggregate with controlled water and pigment only after baseline approval. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

| Trial field | Entry |
|---|---|
| Cement type | CEM class to be recorded by supplier batch |
| Aggregate blend | sand, stone dust and coarse aggregate proportions by weight |
| Water addition | metered water plus moisture adjustment |
| Admixture | only after baseline mix is proven |
| Cube/testing plan | minimum 7-day and 28-day checks; retain batch samples |
| QC focus | prioritize surface finish, abrasion resistance, interlock fit and color consistency |
| Production sign-off | plant manager, QC lead and maintenance lead |

# Mix Design Trial Sheet: Premium and Permeable Pavers

Products covered: cobblestone and grass/permeable pavers. Starting mix basis: high-strength paver mix or gap-graded permeable mix based on void and drainage target. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

| Trial field | Entry |
|---|---|
| Cement type | CEM class to be recorded by supplier batch |
| Aggregate blend | sand, stone dust and coarse aggregate proportions by weight |
| Water addition | metered water plus moisture adjustment |
| Admixture | only after baseline mix is proven |
| Cube/testing plan | minimum 7-day and 28-day checks; retain batch samples |
| QC focus | prioritize void control, edge durability, installed appearance and customer installation guidance |
| Production sign-off | plant manager, QC lead and maintenance lead |

# Mix Design Trial Sheet: Civil Precast Products

Products covered: kerbstones, drainage channels and culverts. Starting mix basis: M25-M35+ design mixes with reinforcement where specified. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

| Trial field | Entry |
|---|---|
| Cement type | CEM class to be recorded by supplier batch |
| Aggregate blend | sand, stone dust and coarse aggregate proportions by weight |
| Water addition | metered water plus moisture adjustment |
| Admixture | only after baseline mix is proven |
| Cube/testing plan | minimum 7-day and 28-day checks; retain batch samples |
| QC focus | prioritize cage placement, cover, compaction, curing and dimensional checks |
| Production sign-off | plant manager, QC lead and maintenance lead |

# Mix Design Trial Sheet: Ready-Mix Concrete

Products covered: M15-M40 supplied by cubic metre. Starting mix basis: engineer-approved design mix by grade, slump and aggregate size. This should be trialed across at least three moisture conditions and two cement sources before production sign-off.

The acceptance gate should combine machine behavior, surface finish, demolding quality, curing result, dimensional tolerance and compressive strength. A ratio that looks cheaper but increases breakage is not acceptable.

| Trial field | Entry |
|---|---|
| Cement type | CEM class to be recorded by supplier batch |
| Aggregate blend | sand, stone dust and coarse aggregate proportions by weight |
| Water addition | metered water plus moisture adjustment |
| Admixture | only after baseline mix is proven |
| Cube/testing plan | minimum 7-day and 28-day checks; retain batch samples |
| QC focus | prioritize batch tickets, slump, temperature where relevant, site sign-off and cube records |
| Production sign-off | plant manager, QC lead and maintenance lead |

# Ready-Mix Concrete Design Families

Ready-mix should be sold by grade, slump and application. Each design family should have a locked recipe, permitted adjustment range, batch ticket and site acceptance protocol.

| Grade | Typical applications | Design focus | QC checks |
|---|---|---|---|
| M15 | blinding, non-structural works | workability and cost control | slump, batch ticket |
| M20 | light slabs and foundations | strength consistency | slump and cubes |
| M25 | commercial slabs and beams | cement efficiency and placement | slump, cubes, temperature |
| M30+ | specified structures | engineering approval | formal mix approval and cube program |

**All RMC mixes must be confirmed by a qualified engineer and site-specific specifications before sale.**

# 6. Raw Materials

**The procurement strategy should use dual sourcing where practical, moisture-controlled storage, supplier scorecards and batch traceability from receipt to dispatch.**

CRITICAL INPUT
## Cement
price, quality and availability dominate gross margin

HIDDEN VARIABLE
## Moisture
sand and dust moisture can undermine recipes

STOCK POLICY
## 14-30 days
depends on product and cement price volatility

Raw material control is the biggest operational determinant of quality and margin. Cement, aggregates, sand, water, steel, pigments and admixtures must be managed as production inputs, not just purchases.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Raw Material Control Register

Every raw material should have a purchase specification, receipt inspection, storage rule and rejection procedure. A cheap material that creates rejects is a margin leak.

| Input | Specification | Storage/control | Risk |
|---|---|---|---|
| Cement | CEM I/II or grade required by product | silo and bag storage; FIFO; certificates | price volatility and supply interruption |
| Washed sand | clean, graded, low clay/silt | covered bays and moisture checks | strength loss and surface defects |
| Stone dust | consistent grading | separate from sand; avoid contamination | water demand and dimensional variation |
| Coarse aggregate | 6-20 mm as mix requires | segregated bins; loader discipline | honeycombing and poor compaction |
| Water | potable or approved production water | tank, meter, recycling controls | strength and efflorescence issues |
| Steel reinforcement | certified bars/wire cages | dry storage and certificates | culvert/channel structural risk |
| Pigments/admixtures | approved supplier and dosage | locked dosing and trial records | color mismatch and setting variability |
| Pallets/molds release | machine-compatible consumables | maintenance register | surface defects and downtime |

# Material Sheet: Cement

Specification focus: CEM I/II or grade required by product. Control requirement: silo and bag storage; FIFO; certificates. Key risk: price volatility and supply interruption.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | silo and bag storage; FIFO; certificates |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Washed sand

Specification focus: clean, graded, low clay/silt. Control requirement: covered bays and moisture checks. Key risk: strength loss and surface defects.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | covered bays and moisture checks |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Stone dust

Specification focus: consistent grading. Control requirement: separate from sand; avoid contamination. Key risk: water demand and dimensional variation.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | separate from sand; avoid contamination |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Coarse aggregate

Specification focus: 6-20 mm as mix requires. Control requirement: segregated bins; loader discipline. Key risk: honeycombing and poor compaction.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
| --- | --- |
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | segregated bins; loader discipline |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Water

Specification focus: potable or approved production water. Control requirement: tank, meter, recycling controls. Key risk: strength and efflorescence issues.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | tank, meter, recycling controls |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Steel reinforcement

Specification focus: certified bars/wire cages. Control requirement: dry storage and certificates. Key risk: culvert/channel structural risk.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | dry storage and certificates |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Pigments/admixtures

Specification focus: approved supplier and dosage. Control requirement: locked dosing and trial records. Key risk: color mismatch and setting variability.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
|---|---|
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | locked dosing and trial records |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Material Sheet: Pallets/molds release

Specification focus: machine-compatible consumables. Control requirement: maintenance register. Key risk: surface defects and downtime.

Procurement should record supplier, delivery date, batch or truck reference, moisture/visual inspection and whether the material is approved, quarantined or rejected.

| Control point | Procedure |
| --- | --- |
| Supplier qualification | collect certificates, route reliability and delivered-cost history |
| Receipt inspection | visual check, weight/volume reconciliation and contamination check |
| Storage | maintenance register |
| Production release | QC or supervisor approval before use |
| Monthly scorecard | price, delivery reliability, rejection rate and complaint linkage |

# Supply Chain Diagram

The factory should minimize double handling. Aggregates and sand flow to covered bays, cement to silos or dry stores, water through metering and recycling, and finished products through curing to yard zones.

**Raw material to finished goods control flow**

Supplier → Receipt QC → Storage bay → Batching → Production → Curing → Dispatch

# 7. Machinery Options

**The investment should avoid under-buying the core line and then losing margin through poor consistency, high rejects and downtime. However, ready-mix and culvert capacity should be phased by customer evidence.**

RECOMMENDED

## Option B

automatic vibropress core line

KEY PROCUREMENT RISK

## Support

local service, spares and commissioning discipline

MOLD STRATEGY

## modular

blocks, pavers, kerbs and premium profiles

Machinery selection determines quality, labor intensity, mold flexibility, downtime risk and expansion potential. The recommended base case is a modular automatic line with local service planning and a realistic spares pack.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Machinery Option Comparison

Option B is the recommended base-case line because it offers credible quality and capacity without overcommitting to a national-scale plant before demand is proven.

| Option | Description | Indicative CAPEX | Pros | Cons |
|---|---|---|---|---|
| A | Semi-automatic block/paver line | UGX 5.80 bn | lower entry cost; easier staffing | higher labor, lower consistency, slower scale |
| B | Automatic vibropress with batching and stacker | UGX 9.40 bn | recommended quality and throughput | higher funding and maintenance discipline |
| C | High-capacity integrated plant with robotics | UGX 15.50 bn | best scale economics | too much demand risk before anchor contracts |
| D | Used/refurbished imported line | UGX 4.90 bn | lower headline cost | hidden downtime, spares and controls risk |

# Core Vibropress Line

Scope: automatic block/paver machine, board feed, stacker and mold set. Investor value: dimension consistency, fast mold changeover and vibration control.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
|---|---|
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Batching and Mixing

Scope: aggregate bins, cement weighing, pan mixer and water meter. Investor value: recipe repeatability and moisture correction.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
|---|---|
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

## Curing System

Scope: covered racks, water misting, chambers and product lot control. Investor value: strength development and reduced breakage.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
|---|---|
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# RMC Plant

Scope: 30-60 m3/h batching plant, silos, controls and transit mixers. Investor value: commercial concrete supply and site pours.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
| --- | --- |
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Culvert and Drainage Equipment

Scope: pipe molds, cage rollers, vibrators, cranes and curing area. Investor value: road/drainage tender capability.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
|---|---|
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Material Handling

Scope: wheel loader, forklifts, pallet trucks, tippers and flatbeds. Investor value: yard productivity and delivery reliability.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
| --- | --- |
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Power Backup

Scope: generator, transfer switch and critical load plan. Investor value: reduced downtime from outages.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
| --- | --- |
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Weighbridge and ERP

Scope: truck weighbridge, batch tickets, inventory and invoicing. Investor value: shrinkage control and finance visibility.

Procurement should require installation drawings, spare parts list, warranty terms, training schedule, acceptance tests and written throughput assumptions by product family.

| Procurement requirement | Reason |
|---|---|
| Factory acceptance test | confirms machine function before shipment |
| Site acceptance test | confirms installed throughput and product quality |
| Critical spares pack | protects first-year uptime |
| Operator training | reduces rejects and maintenance abuse |
| Mold documentation | ensures dimensions, wear limits and replacement planning |

# Machinery Procurement Gate

No equipment order should be released until civil foundations, power requirements, compressed air, water pressure, installation sequence and commissioning responsibility are frozen.

**Machinery procurement and commissioning gate**

**Technical spec** → **Supplier bids** → **FAT** → **Shipping** → **Civil readiness** → **Install** → **SAT** → **Operator sign-off**

**Investor gate: retain final supplier payment until site acceptance test confirms products, throughput and controls.**

# 8. Factory Layout

**The recommended layout is a one-way material flow with segregated heavy-vehicle movement, covered curing, laboratory adjacency to production and finished goods organized by product family and delivery zone.**

| SITE NEED | PRIMARY FLOW | LAYOUT PRIORITY |
| --- | --- | --- |
| **5-8 acres** | **one-way** | **yard control** |
| depending on stockyard depth and RMC fleet | raw material to dispatch without cross-traffic | stock, curing and dispatch discipline |

The site layout must protect flow: raw materials in, controlled production, curing, quality release, finished goods storage and dispatch out. Poor layout will consume margin through double handling.

**1 Investor lens**

What the chapter means for return, risk and bankability.

**2 Operating lens**

Factory routines, people, quality and throughput implications.

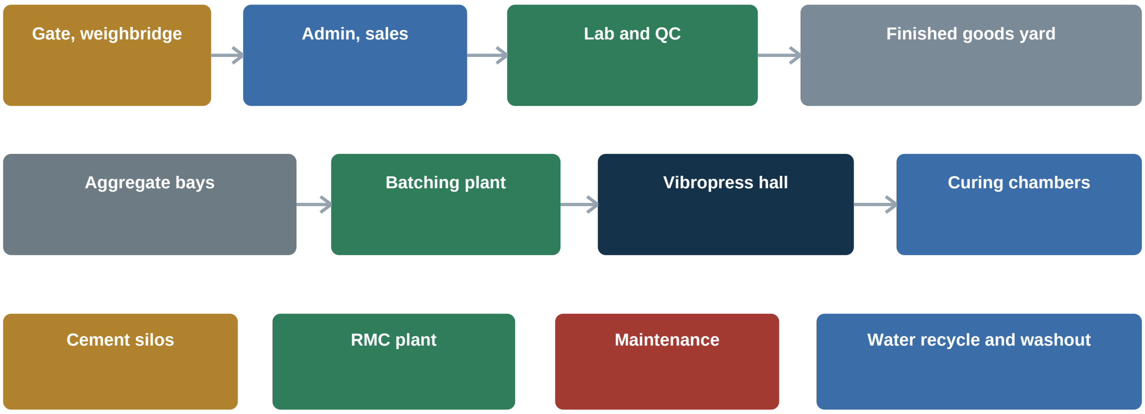
**3 Decision lens**

Actions required before financial close and procurement.

## Indicative Factory Layout

The diagram is conceptual and unscaled. The final layout should be completed by an industrial engineer after geotechnical results, drainage design, truck turning radii and utility routes are known.

Indicative unscaled layout: segregated raw material, production, curing, laboratory and dispatch movements.



# Layout Zone: Gate and Weighbridge

Control all inbound raw materials and outbound product loads; capture truck, supplier, product, weight and document number.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
|---|---|
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Aggregate Bays

Use separated covered bays to avoid contamination and allow moisture testing by material type.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
|---|---|
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Cement Silos and Bag Store

Protect cement from moisture; design access for bulk tanker and bag deliveries without blocking production.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
| --- | --- |
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Production Hall

Place batching, mixer, vibropress and pallet flow in a linear sequence with maintenance access.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
|---|---|
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Curing Zone

Separate green products from released stock; maintain curing logs and water/mist controls.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
|---|---|
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Finished Goods Yard

Organize by product, age, lot, customer reservation and dispatch route.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
|---|---|
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# Layout Zone: Ready-Mix Bay

Keep RMC traffic from block dispatch queues; include washout and water recycling controls.

Design principle: vehicle movement, people movement and material movement should be visible, signed and separated where practical. The layout should make the correct operating behavior the easiest behavior.

| Design check | Requirement |
| --- | --- |
| Safety | clear walkways, PPE zones, speed limits and reversing controls |
| Product integrity | no mixing of green, curing and released product |
| Inventory | lot labels, aging status and stock count access |
| Drainage | falls away from production; sediment and washout management |

# 9. Electricity and Water Consumption

**Uganda's distribution landscape changed when UEDCL assumed electricity distribution operations from Umeme in April 2025. The project should confirm connection capacity, transformer needs and reliability directly with the current distribution utility and ERA guidance.**

POWER DEMAND

**450-750 kVA**

indicative installed load range by phase

WATER DEMAND

**45-125 m3/day**

depends on RMC scale, curing and recycling

CRITICAL CONTROL

**metering**

water, cement and power by batch/shift

Electricity and water are production risks, not back-office utilities. The plant requires reliable industrial power, backup for critical operations, metered water, recycling and batch-level water control.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Utility Demand Model

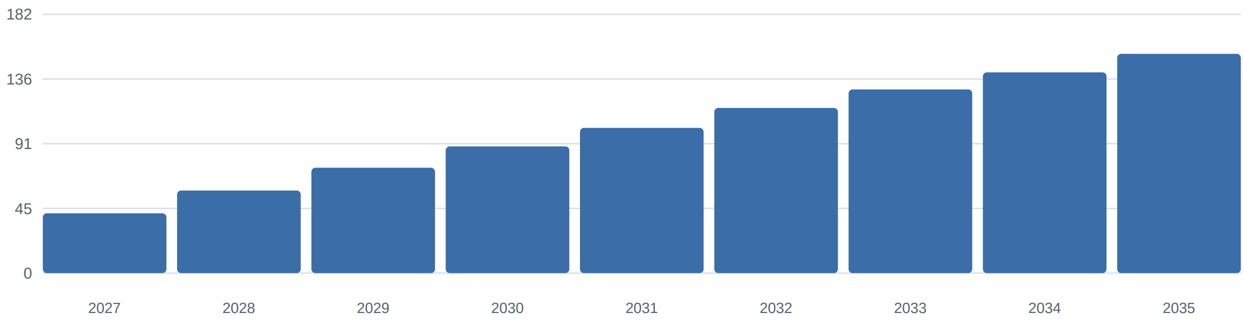
The estimates below are planning figures. Electrical engineering must confirm installed load, diversity factor, starting currents, transformer size, backup generator rating and power factor correction.

| Area | Installed load | Typical operating use | Control action |
|---|---|---|---|
| Vibropress and batching | 180-280 kW | 80-160 kWh/hour in production | shift energy dashboard |
| RMC plant and pumps | 80-150 kW | 35-90 kWh/hour during pours | separate RMC meter |
| Curing, water pumps and compressors | 40-90 kW | variable by curing method | pressure and water meter |
| Lighting/admin/security | 25-45 kW | low but continuous | LED and submeter |
| Workshop and auxiliaries | 50-120 kW | maintenance peaks | planned maintenance windows |

# Electricity Consumption Ramp

The base model assumes electricity intensity improves as utilization rises. The absolute bill increases with volume, but kWh per saleable unit should fall as stoppages and rejects reduce.

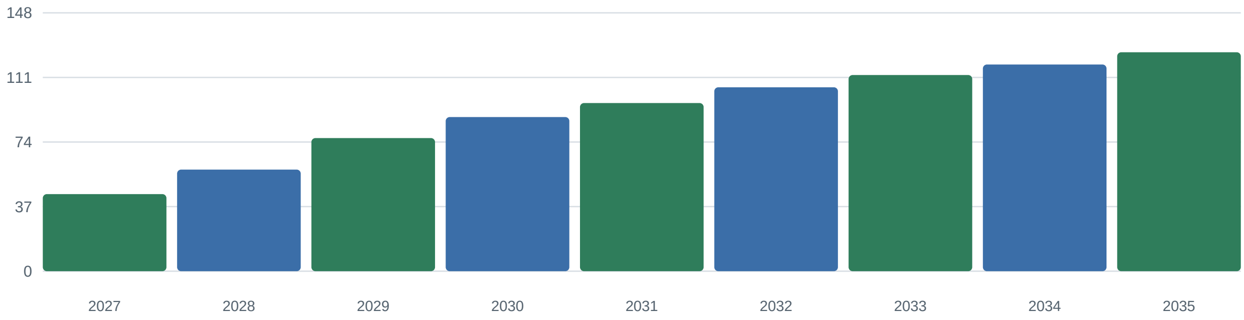
**Indicative monthly electricity use by maturity stage, MWh**



# Water Consumption Ramp

Water use comes from mixing, curing, dust control, washout, sanitation and ready-mix production. Recycling is especially important after RMC launch because washout water and sediment can become environmental and cost risks.

**Indicative average production water use, m3/day**



# UEDCL Connection Plan

Submit load schedule, transformer requirement, single-line diagram and site plan; confirm tariff category and service reliability expectations.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

| KPI | Target direction |
|---|---|
| kWh per saleable unit | down as utilization and maintenance improve |
| Generator hours | low, exception-based |
| Water recycled | up with RMC and curing maturity |
| Unplanned utility stoppages | down with preventive maintenance |

# Backup Power Strategy

Back up controls, batching completion, lighting, lab and critical pumps; avoid running the entire plant on generator unless economics justify it.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

| KPI | Target direction |
|---|---|
| kWh per saleable unit | down as utilization and maintenance improve |
| Generator hours | low, exception-based |
| Water recycled | up with RMC and curing maturity |
| Unplanned utility stoppages | down with preventive maintenance |

# Water Source Strategy

Use NWSC where available, evaluate borehole/rainwater only with permits and water quality testing, and install storage tanks for operating buffer.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

| KPI | Target direction |
|---|---|
| kWh per saleable unit | down as utilization and maintenance improve |
| Generator hours | low, exception-based |
| Water recycled | up with RMC and curing maturity |
| Unplanned utility stoppages | down with preventive maintenance |

# Water Recycling and Washout

Separate washout, sediment traps and clarified reuse water; protect drains and maintain a visible environmental housekeeping routine.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

| KPI | Target direction |
|---|---|
| kWh per saleable unit | down as utilization and maintenance improve |
| Generator hours | low, exception-based |
| Water recycled | up with RMC and curing maturity |
| Unplanned utility stoppages | down with preventive maintenance |

# Utility KPIs

Track kWh per unit, m3 water per batch, downtime by cause, generator hours, power factor and water recycling percentage.

Utilities should be reviewed in the weekly management meeting because they affect cost, uptime, compliance and product quality.

| KPI | Target direction |
|---|---|
| kWh per saleable unit | down as utilization and maintenance improve |
| Generator hours | low, exception-based |
| Water recycled | up with RMC and curing maturity |
| Unplanned utility stoppages | down with preventive maintenance |

# 10. Laboratory and Quality Control

**The QC system should cover raw materials, mix recipes, machine settings, curing, dimensions, compressive strength, visual defects, release status and complaint investigation.**

CUBE PROGRAM

## 7 and 28 days

strength development tracking

RELEASE GATE

## QC sign-off

no green stock dispatched

TRACEABILITY

## lot number

batch ticket to customer delivery

The laboratory is the project's trust engine. It turns concrete products from commodity blocks into documented, traceable and certifiable building materials.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Quality Control System

NBRB guidance identifies concrete cube testing, curing and mechanical compaction as quality-control fundamentals. The factory should formalize these into daily routines with records that can be shown to customers and inspectors.

### Factory QC control loop

Raw material check → Recipe batch → Machine settings → Green product check → Curing log → Cube test → Release → Complaint feedback

# Laboratory Equipment Register

The laboratory should be modest but credible at launch. A clean, calibrated and used lab is more valuable than an expensive lab that production ignores.

| Equipment | Purpose | Control |
|---|---|---|
| Compression testing machine | cube strength verification | calibration certificate and safety guard |
| Cube molds and curing tank | 7-day and 28-day testing | mold cleaning, labeling and curing water control |
| Sieve set and shaker | aggregate grading | weekly grading logs by supplier |
| Moisture meter | sand and aggregate moisture correction | shift-level moisture entries |
| Slump cone and rod | RMC workability | site acceptance and batch ticket match |
| Digital scales | weigh-batch control and lab trials | calibration and check weights |
| Calipers/tape/gauges | dimensional tolerance | product release checks |
| Sample retain area | dispute resolution | retained samples by lot and age |

# Incoming Material Inspection

Inspect cement certificates, sand cleanliness, aggregate grading, moisture, contamination and delivery weights before release to production.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# Batch Ticket Discipline

Each production batch should record recipe, operator, machine, material lots, water adjustment, start time and product family.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# Dimensional Tolerance Checks

Measure units at defined intervals. Reject trends usually point to mold wear, poor vibration, wrong moisture or handling damage.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# Curing Compliance

Log curing start, product age, water/mist conditions and release date. Green products must be physically segregated.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# Compressive Strength Testing

Use cube tests for design validation and routine control. Trend results by mix family and cement source.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# RMC Site Acceptance

For ready-mix, delivery tickets, slump results, time at site, water addition rules and sign-off protect both factory and customer.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
| --- | --- |
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# Complaint Investigation

Every complaint should trace product, batch, customer, delivery, photos, test records and corrective action.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
| --- | --- |
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# UNBS Certification Preparation

Maintain product specifications, test results, process controls and management responsibility for certification visits.

Quality records should be designed so a sales manager can produce customer-ready evidence without disrupting the lab.

| Record | Minimum fields |
|---|---|
| Owner | QC lead or delegated technician |
| Frequency | defined by product risk and production volume |
| Escalation | plant manager if result is out of tolerance |
| Retention | digital and hard-copy files by month and product |

# 11. Legal and Regulatory Offices

**The compliance strategy should be a named approvals register with owners, documents, status, expiry dates and renewal reminders. Regulatory diligence should begin before land is committed.**

COMPANY

## URSB

registration and legal identity

ENVIRONMENT

## NEMA

project brief or ESIA route

PRODUCT

## UNBS

standards and certification pathway

The project touches company registration, investment licensing, tax, environmental assessment, building control, product standards, labor, utilities and local trade licensing.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Regulatory Office Map

The approval path should be managed as a project workstream. Missing approvals can delay commissioning, expose the sponsor to penalties and weaken lender confidence.

| Office | Role | Documents/actions |
|---|---|---|
| URSB | company registration, legal documents | incorporation, beneficial ownership, filings |
| URA | TIN, VAT, PAYE, income tax | tax registration, EFRIS, returns, compliance certificate |
| UIA | investment license and facilitation | license application, land facilitation, aftercare |
| Mbarara City Council | trade license, physical planning, building approvals | zoning, building plans, local permits |
| NEMA | environmental screening, project brief/ESIA | ELMIS submission, approval conditions, audits |
| UNBS | product standards and certification | precast paving blocks, cement/product compliance |
| NBRB / Building Control | building code and construction oversight | factory building plans and inspections |
| UEDCL / ERA | electricity connection and distribution context | load application, metering, tariff and supply reliability |
| NWSC / Water authorities | water and sewerage connection | connection documents, meters and discharge arrangements |
| NSSF and labor/OHS offices | employee social security and workplace safety | registration, payroll compliance, safety records |

# Regulatory Sheet: URSB

Primary role: company registration, legal documents. Key documents/actions: incorporation, beneficial ownership, filings.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | URSB |
| Approval or compliance action | company registration, legal documents |
| Core file | incorporation, beneficial ownership, filings |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: URA

Primary role: TIN, VAT, PAYE, income tax. Key documents/actions: tax registration, EFRIS, returns, compliance certificate.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | URA |
| Approval or compliance action | TIN, VAT, PAYE, income tax |
| Core file | tax registration, EFRIS, returns, compliance certificate |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: UIA

Primary role: investment license and facilitation. Key documents/actions: license application, land facilitation, aftercare.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | UIA |
| Approval or compliance action | investment license and facilitation |
| Core file | license application, land facilitation, aftercare |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: Mbarara City Council

Primary role: trade license, physical planning, building approvals. Key documents/actions: zoning, building plans, local permits.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
| --- | --- |
| Authority | Mbarara City Council |
| Approval or compliance action | trade license, physical planning, building approvals |
| Core file | zoning, building plans, local permits |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: NEMA

Primary role: environmental screening, project brief/ESIA. Key documents/actions: ELMIS submission, approval conditions, audits.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | NEMA |
| Approval or compliance action | environmental screening, project brief/ESIA |
| Core file | ELMIS submission, approval conditions, audits |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: UNBS

Primary role: product standards and certification. Key documents/actions: precast paving blocks, cement/product compliance.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | UNBS |
| Approval or compliance action | product standards and certification |
| Core file | precast paving blocks, cement/product compliance |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: NBRB / Building Control

Primary role: building code and construction oversight. Key documents/actions: factory building plans and inspections.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | NBRB / Building Control |
| Approval or compliance action | building code and construction oversight |
| Core file | factory building plans and inspections |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: UEDCL / ERA

Primary role: electricity connection and distribution context. Key documents/actions: load application, metering, tariff and supply reliability.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | UEDCL / ERA |
| Approval or compliance action | electricity connection and distribution context |
| Core file | load application, metering, tariff and supply reliability |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: NWSC / Water authorities

Primary role: water and sewerage connection. Key documents/actions: connection documents, meters and discharge arrangements.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | NWSC / Water authorities |
| Approval or compliance action | water and sewerage connection |
| Core file | connection documents, meters and discharge arrangements |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Regulatory Sheet: NSSF and labor/OHS offices

Primary role: employee social security and workplace safety. Key documents/actions: registration, payroll compliance, safety records.

Assign one internal owner and one external advisor where appropriate. Record submission date, officer contact, statutory timeline, comments, approval conditions and renewal date.

| Register field | Entry |
|---|---|
| Authority | NSSF and labor/OHS offices |
| Approval or compliance action | employee social security and workplace safety |
| Core file | registration, payroll compliance, safety records |
| Responsible owner | project manager during setup; compliance officer after launch |
| Board visibility | monthly until commissioning; quarterly after stabilization |

# Compliance Calendar

The compliance calendar should be reviewed monthly. Tax and payroll failures can quickly become investor issues even when production is performing well.

| Frequency | Compliance action | Owner |
|---|---|---|
| Monthly | PAYE, NSSF, VAT returns where applicable, utility bills, safety toolbox records | Finance / HR / HSE |
| Quarterly | management accounts, board compliance report, supplier certificates | Finance / QC |
| Semi-annual | environmental housekeeping review, emergency drills | HSE / Plant manager |
| Annual | company filings, trade license renewal, insurance, product certification review | Compliance officer |
| As required | NEMA approval conditions, building inspections, standards audits | Project manager / QC |

# 12. CAPEX and OPEX

**The base case funding envelope is large enough to avoid a common manufacturing failure mode: buying the machine but underfunding power, yard, curing, fleet, lab and cash cycle.**

| TOTAL FUNDING | CAPEX SUBTOTAL | CONTINGENCY |
|---|---|---|
| **UGX 36.13 bn** | **UGX 30.31 bn** | **8%** |
| including contingency and opening working capital | before contingency and working capital | applied to CAPEX subtotal |

The project requires a funding envelope that covers not just machinery, but site works, civil works, utilities, fleet, laboratory, pre-opening costs, contingency and working capital.

1. **Investor lens**
   What the chapter means for return, risk and bankability.
2. **Operating lens**
   Factory routines, people, quality and throughput implications.
3. **Decision lens**
   Actions required before financial close and procurement.

## CAPEX Schedule

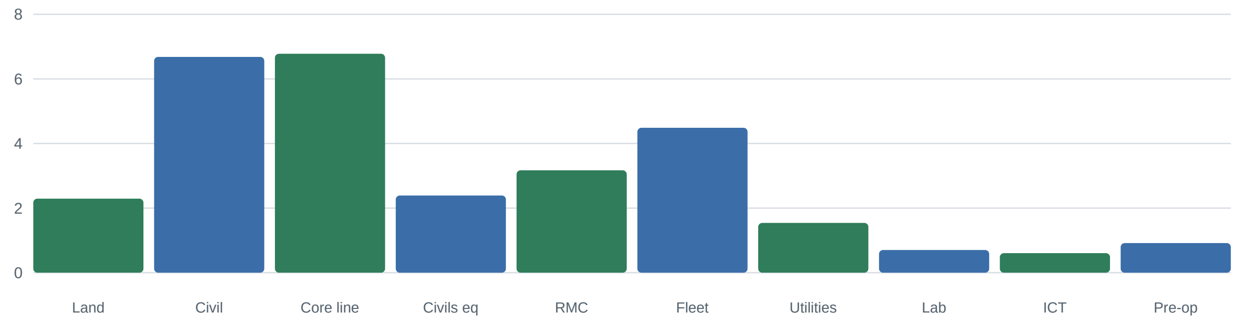
The schedule is a planning estimate. Supplier quotations and a local quantity surveyor should replace these figures before financing.

| CAPEX item | Amount | Scope note |
|---|---|---|
| Land lease premium, surveys and site enabling | UGX 2,350,000,000 | 5-8 acre industrial plot, drainage, fencing and ground preparation |
| Civil works and factory buildings | UGX 6,850,000,000 | production hall, curing sheds, stockyard pavement, admin and lab |
| Block and paver vibropress line | UGX 6,950,000,000 | automatic batching, mixer, block machine, molds, stacker and pallets |
| Kerb, channel and culvert production equipment | UGX 2,450,000,000 | forms, cage rollers, pipe molds, cranes and vibration systems |
| Ready-mix plant and concrete pumps | UGX 3,250,000,000 | 30-60 m3 per hour batching plant and dispatch controls |
| Mobile plant and logistics fleet | UGX 4,600,000,000 | wheel loader, forklifts, tipper, flatbeds, transit mixers and pickup |
| Power, water and utilities infrastructure | UGX 1,580,000,000 | transformer, wiring, borehole/NWSC interface, tanks and recycling |
| Laboratory and quality control equipment | UGX 720,000,000 | cube press, molds, sieves, scales, slump kits and curing tanks |
| ERP, weighbridge, security and ICT | UGX 620,000,000 | dispatch control, accounting, CCTV, access control and weighbridge |
| Pre-opening costs and training | UGX 940,000,000 | technical commissioning, launch marketing, permits and operator training |
| Contingency | UGX 2,424,800,000 | 8% planning contingency |
| Opening working capital | UGX 3,400,000,000 | cement, aggregate, payroll, spares and receivables buffer |
| Total funding envelope | UGX 36,134,800,000 | base-case financing need |

# CAPEX Visual

Machinery, civil works and fleet dominate the investment. Underfunding the yard and utilities would directly reduce factory performance.

**CAPEX by category, UGX bn**



# OPEX Model

OPEX discipline depends on three routines: batch yield, maintenance uptime and cash collection. Management should review each weekly.

| OPEX category | Year 2027 | Mature year | Management control |
| --- | --- | --- | --- |
| Cement and aggregates | largest variable cost | largest variable cost | supplier contracts, moisture control, waste reduction |
| Labor and payroll | UGX 2.15 bn | UGX 4.78 bn | shift planning and productivity per employee |
| Utilities | UGX 0.58 bn | UGX 1.64 bn | submetering, power factor and water recycling |
| Maintenance | UGX 0.42 bn | UGX 1.61 bn | preventive maintenance and spares planning |
| Sales/admin/logistics overhead | UGX 1.05 bn | UGX 3.14 bn | route economics and credit control |

# Cement Cost Sensitivity

A cement price increase flows directly into gross margin unless sales prices, mix efficiency or procurement terms adjust.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Aggregate and Sand Cost Sensitivity

Delivered aggregate cost depends on quarry distance, truck utilization, moisture and rejection rates.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
| --- | --- |
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Labor Productivity

Labor should be measured per saleable unit, not headcount alone. Automation only pays off when operators, maintenance and QC routines work together.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Maintenance Budget

Preventive maintenance protects production days. The budget should include molds, wear parts, hydraulic components, vibration motors and critical sensors.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Fleet and Distribution Cost

Freight must be priced by zone. A product can be profitable ex-works and loss-making after poor route planning.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

## Working Capital

Receivables, cement stock, pallets and curing inventory can absorb cash faster than accounting profit suggests.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Insurance and Security

The plant should carry property, machinery breakdown, vehicle, liability, workers and goods-in-transit cover.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# Management Controls

ERP, weighbridge, stock counts and invoice matching protect against shrinkage and unbilled deliveries.

Investor diligence should require monthly variance reporting by product family and reason codes for material, labor, downtime and price variance.

| Control | Practical action |
|---|---|
| Budget owner | assign named manager |
| Variance threshold | investigate exceptions above management tolerance |
| Frequency | weekly for production, monthly for board reporting |
| Evidence | ERP, batch tickets, stock counts and supplier invoices |

# 13. 10-Year Financial Forecast

**This model is an investor screening case. It should be rebuilt in spreadsheet form with supplier quotes, financing terms, tax advice and signed customer pipeline before investment approval.**

FORECAST PERIOD
**2026-2035**
10-year horizon requested

COMMERCIAL LAUNCH
**2027**
first full operating year

CURRENCY
**UGX only**
all figures in nominal UGX

The 2026-2035 model shows a setup year followed by staged revenue growth, improving gross margin, disciplined OPEX and meaningful cash generation after ramp-up.

1 **Investor lens**
What the chapter means for return, risk and bankability.

2 **Operating lens**
Factory routines, people, quality and throughput implications.

3 **Decision lens**
Actions required before financial close and procurement.

## Forecast Summary Table

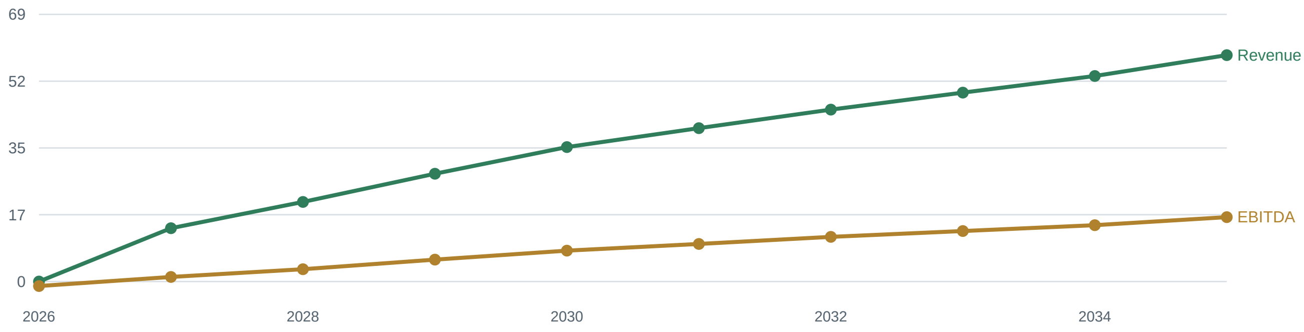
The unlevered free cash flow line treats 2026 as the investment year and excludes debt service. Financing structure will change shareholder returns and lender coverage metrics.

| Year | Revenue | Gross profit | EBITDA | EBITDA margin | Tax | Unlevered FCF |
|---|---|---|---|---|---|---|
| 2026 | UGX 0.00 bn | UGX 0.00 bn | UGX -1.17 bn | 0.0% | UGX 0.00 bn | UGX -37.30 bn |
| 2027 | UGX 13.80 bn | UGX 5.38 bn | UGX 1.18 bn | 8.6% | UGX 0.00 bn | UGX 1.00 bn |
| 2028 | UGX 20.60 bn | UGX 8.45 bn | UGX 3.19 bn | 15.5% | UGX 0.22 bn | UGX 2.75 bn |
| 2029 | UGX 27.90 bn | UGX 12.00 bn | UGX 5.67 bn | 20.3% | UGX 0.97 bn | UGX 4.42 bn |
| 2030 | UGX 34.80 bn | UGX 15.31 bn | UGX 7.99 bn | 23.0% | UGX 1.66 bn | UGX 5.98 bn |
| 2031 | UGX 39.70 bn | UGX 17.86 bn | UGX 9.72 bn | 24.5% | UGX 2.18 bn | UGX 7.08 bn |
| 2032 | UGX 44.50 bn | UGX 20.47 bn | UGX 11.56 bn | 26.0% | UGX 2.73 bn | UGX 8.25 bn |
| 2033 | UGX 48.90 bn | UGX 22.74 bn | UGX 13.07 bn | 26.7% | UGX 3.19 bn | UGX 9.16 bn |
| 2034 | UGX 53.20 bn | UGX 25.00 bn | UGX 14.58 bn | 27.4% | UGX 3.64 bn | UGX 10.06 bn |
| 2035 | UGX 58.60 bn | UGX 27.84 bn | UGX 16.66 bn | 28.4% | UGX 4.26 bn | UGX 11.35 bn |

# Revenue and EBITDA Trend

The model depends on ramp-up. If capacity utilization grows slower than planned, management must protect cash by staging product launches and controlling credit.

**Revenue and EBITDA, UGX bn**



# Revenue Mix, UGX bn

Blocks dominate early utilization; ready-mix and civil products increase as the factory builds contractor credibility and fleet capability.

| Year | Blocks | Pavers | Civils | Ready-mix | Total |
|---|---|---|---|---|---|
| 2026 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2027 | 5.80 | 4.14 | 2.21 | 1.66 | 13.80 |
| 2028 | 8.65 | 6.18 | 3.30 | 2.47 | 20.60 |
| 2029 | 11.72 | 8.37 | 4.46 | 3.35 | 27.90 |
| 2030 | 12.18 | 9.74 | 6.26 | 6.61 | 34.80 |
| 2031 | 13.89 | 11.12 | 7.15 | 7.54 | 39.70 |
| 2032 | 15.57 | 12.46 | 8.01 | 8.46 | 44.50 |
| 2033 | 17.11 | 13.69 | 8.80 | 9.29 | 48.90 |
| 2034 | 18.62 | 14.90 | 9.58 | 10.11 | 53.20 |
| 2035 | 20.51 | 16.41 | 10.55 | 11.13 | 58.60 |

# Financial Sensitivity

The most important investor sensitivities are utilization, cement cost, receivable days and product mix. These should become lender-case sensitivities in the next model version.

| Scenario | Change | Impact | Management response |
|---|---|---|---|
| Base case | as modeled | positive NPV and mature EBITDA margin above 20% | execute staged launch |
| Slow ramp | revenue 15% lower in 2027-2029 | payback moves later and working capital tightens | defer RMC fleet and preserve cash |
| Cement shock | cement cost 12% higher | gross margin pressure in blocks and pavers | price escalation, recipe efficiency and supplier split |
| Receivable stress | DSO +30 days | funding need increases materially | tighten credit and deposits |
| Premium upside | paver/civils mix +10% | margin improves and freight economics improve | market premium profiles and municipal contracts |

# Assumption: Utilization

The model assumes the core line ramps from single-shift launch to higher utilization by product family. Utilization is the main break-even driver.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Assumption: Gross Margin

Gross margin improves as rejects reduce, procurement is contracted, premium pavers grow and ready-mix batching becomes disciplined.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Assumption: Payroll

Payroll includes production, QC, maintenance, sales, dispatch, admin and security. Employer statutory costs should be confirmed.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Assumption: Tax

A generic income tax assumption is included for modeling. Final tax treatment, depreciation and incentives require professional advice.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Assumption: Maintenance CAPEX

Maintenance capital expenditure increases as plant utilization rises and molds, pallets and fleet require replacement.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Assumption: Working Capital

Opening working capital covers raw material, payroll, spares, receivables and curing inventory during ramp-up.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Debt Sizing Considerations

Lenders will focus on DSCR, collateral, sponsor equity, offtake quality, construction risk and working-capital controls.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# Equity Return Considerations

Equity upside depends on utilization, disciplined credit, product mix and ability to replicate distribution into regional markets.

The next financial model should be spreadsheet-based with input tabs, working-capital schedule, debt schedule, depreciation schedule and covenant outputs.

| Model item | Base-case treatment |
|---|---|
| Currency | UGX only |
| Inflation | embedded in nominal revenue and cost ramp |
| Financing | unlevered forecast; debt structure not included in cash flow |
| Decision gate | replace assumptions with quotes and customer evidence |

# 14. Risk Register

**Risk ownership should be integrated into weekly management reporting during setup and monthly board reporting after launch.**

RISKS TRACKED

## 15

strategic, operating, market and compliance risks

TOP SEVERITY

## quality/cash

failed product or receivables can erode trust quickly

CONTROL STYLE

## owner-led

each risk must have a named owner and metric

The project has manageable risks if they are owned early. The highest risks are raw material volatility, quality failures, utility reliability, credit control, environmental approvals and commissioning discipline.

1 **Investor lens**

What the chapter means for return, risk and bankability.

2 **Operating lens**

Factory routines, people, quality and throughput implications.

3 **Decision lens**

Actions required before financial close and procurement.

## Risk Register Master Table

The risk register should be maintained as a live operating document. Severity should be recalibrated after site selection, supplier contracting and customer pipeline validation.

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Cement price spike | High | High | Supplier framework agreements, monthly price index, dual sourcing and cement silo stock discipline. |
| Aggregate moisture variation | High | Medium | Moisture testing by shift, covered bays, calibration of water dosing and batch tickets. |
| Power interruption | Medium | High | UEDCL industrial connection, standby generator, maintenance windows and production buffer stock. |
| Weak curing discipline | Medium | High | Curing chambers, QC sign-off before dispatch and batch traceability by product lot. |
| Price undercutting by informal block makers | High | Medium | Segment into certified quality, delivery reliability, credit controls and contractor education. |
| Slow receivables from contractors | High | High | Credit approval, milestone billing, retention caps and early-payment discounts. |
| Environmental approval delays | Medium | High | Early NEMA screening, project brief/ESIA route, wetland setbacks and washout management. |
| Equipment commissioning delay | Medium | High | Supplier FAT, local spares pack, liquidated damages and operator training before shipment. |
| RMC slump or strength disputes | Medium | High | Mix design approvals, cube records, delivery tickets, slump testing and site acceptance protocol. |
| Fleet utilization below plan | Medium | Medium | Dispatch optimization, route costing, backhaul planning and outsourced peak capacity. |
| Foreign exchange exposure on machinery | Medium | Medium | Lock purchase timing, hedge deposits where available, maintain UGX funding contingency. |
| Regulatory non-compliance | Low | High | Compliance calendar covering URSB, URA, NSSF, NEMA, UNBS, city permits and OHS. |
| Safety incident | Medium | High | Traffic separation, LOTO, PPE, lifting plans, near-miss reporting and supervisor accountability. |
| Water shortage or abstraction limit | Medium | Medium | NWSC/borehole redundancy, rainwater harvesting, recycling and water meter dashboard. |
| Expansion execution stretch | Medium | Medium | Stage-gate East African roll-out; do not launch depots before service metrics stabilize. |

# Risk Detail: Cement price spike

Likelihood: High. Impact: High. Mitigation: Supplier framework agreements, monthly price index, dual sourcing and cement silo stock discipline.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
|---|---|
| Risk event | Cement price spike |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Aggregate moisture variation

Likelihood: High. Impact: Medium. Mitigation: Moisture testing by shift, covered bays, calibration of water dosing and batch tickets.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
| --- | --- |
| Risk event | Aggregate moisture variation |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Power interruption

Likelihood: Medium. Impact: High. Mitigation: UEDCL industrial connection, standby generator, maintenance windows and production buffer stock.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
|---|---|
| Risk event | Power interruption |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Weak curing discipline

Likelihood: Medium. Impact: High. Mitigation: Curing chambers, QC sign-off before dispatch and batch traceability by product lot.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
| --- | --- |
| Risk event | Weak curing discipline |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Price undercutting by informal block makers

Likelihood: High. Impact: Medium. Mitigation: Segment into certified quality, delivery reliability, credit controls and contractor education.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
|---|---|
| Risk event | Price undercutting by informal block makers |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Slow receivables from contractors

Likelihood: High. Impact: High. Mitigation: Credit approval, milestone billing, retention caps and early-payment discounts.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
| --- | --- |
| Risk event | Slow receivables from contractors |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Environmental approval delays

Likelihood: Medium. Impact: High. Mitigation: Early NEMA screening, project brief/ESIA route, wetland setbacks and washout management.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
|---|---|
| Risk event | Environmental approval delays |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# Risk Detail: Equipment commissioning delay

Likelihood: Medium. Impact: High. Mitigation: Supplier FAT, local spares pack, liquidated damages and operator training before shipment.

The risk owner should define one early-warning metric and one immediate response action. Risk reporting should be practical enough to change behavior, not just satisfy a board pack.

| Field | Entry |
|---|---|
| Risk event | Equipment commissioning delay |
| Primary owner | to be assigned during project setup |
| Early warning | define measurable trigger before commissioning |
| Control evidence | policy, records, dashboard or inspection proof |
| Board escalation | required when likelihood or impact moves to high |

# 15. Implementation Roadmap 2026-2035

**Each year has a gating logic. The factory should not accelerate expansion if quality, cash conversion or service levels are below thresholds.**

| ROADMAP PERIOD | FIRST GATE | EXPANSION GATE |
|---|---|---|
| **2026-2035** | **2026** | **2033+** |
| matches requested implementation horizon | site, approvals, engineering and financial close | only after domestic proof of execution |

The roadmap is staged to reduce execution risk: approvals and procurement in 2026, core production in 2027, product expansion through 2030, operating maturity through 2032 and regional options from 2033.

**1 Investor lens**

What the chapter means for return, risk and bankability.

**2 Operating lens**

Factory routines, people, quality and throughput implications.

**3 Decision lens**

Actions required before financial close and procurement.

## Roadmap Overview

The roadmap turns a single factory into a disciplined operating platform. The sponsor should approve each stage only when evidence from the prior stage is acceptable.

| Year | Milestone |
|---|---|
| 2026 | Financial close, site acquisition, ESIA route, detailed engineering, procurement shortlist and anchor customer MOUs. |
| 2027 | Commission block/paver line, launch Mbarara sales office, establish QC lab and secure first municipal/institutional contracts. |
| 2028 | Add kerbstone/channel molds, open contractor credit desk and deepen Bushenyi-Ntungamo-Kabale distribution. |
| 2029 | Commission ready-mix plant, deploy transit mixers and target commercial slabs, hospitals, schools and estates. |
| 2030 | Introduce culvert line, strengthen road-infrastructure bid capability and formalize Rwanda/DRC route intelligence. |
| 2031 | Launch certified permeable paver range, improve water recycling and target climate-resilient urban drainage projects. |
| 2032 | Automate pallet handling, expand stockyard, introduce second shift for pavers and standardize dealer scorecards. |
| 2033 | Open first cross-border depot or bonded distribution partner depending on route economics and receivable quality. |
| 2034 | Evaluate second production cell, larger RMC batch capacity and strategic alliance with quarry or aggregate supplier. |
| 2035 | Regional consolidation, product certification refresh, refinance or exit-readiness review and ESG impact reporting. |

# Roadmap Visual

The staged plan protects capital by linking new product launches and regional moves to proven factory controls.

### Implementation stage gates

**2026 approvals** → **2027 launch** → **2028 pavers/civils** → **2029 RMC** → **2030 culverts** → **2031-32 optimize** → **2033-35 expand**

# Roadmap Year: 2026

Financial close, site acquisition, ESIA route, detailed engineering, procurement shortlist and anchor customer MOUs.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
| --- | --- |
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2027

Commission block/paver line, launch Mbarara sales office, establish QC lab and secure first municipal/institutional contracts.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
| --- | --- |
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2028

Add kerbstone/channel molds, open contractor credit desk and deepen Bushenyi-Ntungamo-Kabale distribution.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2029

Commission ready-mix plant, deploy transit mixers and target commercial slabs, hospitals, schools and estates.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2030

Introduce culvert line, strengthen road-infrastructure bid capability and formalize Rwanda/DRC route intelligence.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2031

Launch certified permeable paver range, improve water recycling and target climate-resilient urban drainage projects.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2032

Automate pallet handling, expand stockyard, introduce second shift for pavers and standardize dealer scorecards.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
| --- | --- |
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2033

Open first cross-border depot or bonded distribution partner depending on route economics and receivable quality.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2034

Evaluate second production cell, larger RMC batch capacity and strategic alliance with quarry or aggregate supplier.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# Roadmap Year: 2035

Regional consolidation, product certification refresh, refinance or exit-readiness review and ESG impact reporting.

Board checkpoint: confirm budget, owner, deliverable evidence, risk status and customer traction before moving to the next gate.

| Workstream | Focus |
|---|---|
| Commercial | pipeline, pricing, customer evidence and collections |
| Operations | throughput, quality, maintenance and safety |
| Finance | budget, working capital and variance control |
| Compliance | approvals, renewals and permit conditions |
| People | hiring, training and accountability |

# 16. East African Expansion Plan

**Potential expansion routes include western Uganda, Rwanda-facing routes, DRC-facing routes, northern Tanzania and selected institutional customers. The plan must remain UGX-based for this report.**

| EXPANSION MODE | PRIORITY ROUTE | GATE METRIC |
| --- | --- | --- |
| **depot first** | **Rwanda/DRC corridors** | **cash conversion** |
| avoid premature second factory | after domestic distribution maturity | do not export receivables problems |

Regional expansion should be distribution-led before it is manufacturing-led. The factory should first prove delivery economics, payment discipline and product acceptance across corridors.

1. **Investor lens**
   What the chapter means for return, risk and bankability.
2. **Operating lens**
   Factory routines, people, quality and throughput implications.
3. **Decision lens**
   Actions required before financial close and procurement.

## Regional Expansion Logic

Concrete products are freight-sensitive. Expansion should prioritize higher-value products, reliable dealer partners and pre-sold project loads before standard blocks travel long distances.

| Market | Entry logic | Best-fit products | Gate condition |
|---|---|---|---|
| Western Uganda deepening | direct fleet and dealers | blocks, pavers, kerbs | service level above target |
| Rwanda-facing corridor | partner depot after route tests | pavers, kerbs, premium products | repeat orders and low damage |
| DRC-facing corridor | project-led export trials | culverts, channels, pavers | secured payment and border logistics |
| Northern Tanzania | opportunistic projects | pavers and RMC not favored except local projects | route economics proven |
| Second plant option | only after 2034 review | full portfolio | capacity constraint and positive NPV |

# Export Readiness Gate

Confirm product certification, packaging, palletization, border documents, customer payment terms, insurance and route damage rates.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
|---|---|
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# Depot Model

Start with a leased stock point and local sales partner rather than building a new plant. Track stock turns, damage and receivable days.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
| --- | --- |
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# Rwanda Corridor

Favor premium pavers, kerbs and packaged products where freight can be justified. Avoid low-margin block exports unless project-specific.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
| --- | --- |
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# DRC Corridor

Use project-led shipments with advance payment or secured terms. Culverts and channels may fit road/drainage demand when payment risk is controlled.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
|---|---|
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# Tanzania Link

Treat as opportunistic until a local partner proves demand and route economics.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
| --- | --- |
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# Regional Brand System

Maintain the same batch traceability, QC sheets and delivery documentation across all markets.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
| --- | --- |
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# Second Factory Decision

A second factory should require proven demand, land availability, management depth, positive NPV and a clear raw-material advantage.

Expansion should be stopped or slowed if product damage, border delays or receivable days exceed thresholds.

| Gate | Evidence |
| --- | --- |
| Demand | repeat orders or signed project pipeline |
| Route economics | delivered margin after freight and damage |
| Cash | advance payment, secure credit or insured receivable |
| Operations | no degradation in domestic service levels |

# 17. Appendices

**Appendix material is deliberately practical: assumptions, scorecards, registers, quality forms, due diligence questions and source references.**

APPENDIX TYPE
**templates**
ready to convert into operational controls

EVIDENCE
**sources**
public-source reference list included

USE
**diligence**
board, lender and management follow-up

The appendices provide detailed worksheets and templates that management can convert into live operating documents.

1 **Investor lens**
What the chapter means for return, risk and bankability.

2 **Operating lens**
Factory routines, people, quality and throughput implications.

3 **Decision lens**
Actions required before financial close and procurement.

# Appendix A: Assumption Register

The assumption register should be transferred into the financial model and marked as assumption, quote, contract, policy or actual result.

| Assumption | Base case | Owner | Validation method |
|---|---|---|---|
| Funding envelope | UGX 36.13 bn | Sponsor/CFO | supplier quotes and QS estimate |
| Commercial launch | 2027 | Project manager | commissioning schedule |
| Gross margin maturity | 47.5% | Plant manager/CFO | product-level costing |
| Working capital | UGX 3.40 bn | CFO | receivables and inventory schedule |
| Water demand | 45-125 m3/day | Operations | metering and engineering design |
| Power demand | 450-750 kVA | Electrical engineer | load schedule and utility confirmation |

# Appendix B: Supplier Scorecard

Supplier management should be evidence-based. The scorecard should be reviewed monthly for cement, sand, stone dust, aggregate, fuel, steel, packaging and admixtures.

| Criteria | Weight | Evidence |
|---|---|---|
| Delivered price stability | 25% | invoice trend and escalation notices |
| On-time delivery | 20% | delivery logs |
| Quality pass rate | 25% | receipt inspection and lab results |
| Documentation | 10% | certificates, tax invoices, batch details |
| Commercial terms | 10% | credit, minimum order, payment flexibility |
| Responsiveness | 10% | issue resolution time |

# Appendix C: Customer Credit Scorecard

Credit should be earned, not granted by relationship pressure. The factory should separate sales approval from credit approval.

| Criteria | Score rule |
|---|---|
| Customer identity | registered business, verified directors and site contact |
| Project evidence | BOQ, contract, owner and payment source |
| Payment history | cash first, then small limit, then review |
| Security | purchase order, deposit, guarantee or retention cap |
| Dispute history | delivery sign-off and complaint behavior |
| Limit review | monthly aging and exposure report |

# Appendix D: Daily Production Report

A simple daily report protects the investor from hidden underperformance. It should be signed by production, QC and dispatch.

| Field | Entry |
| --- | --- |
| Date / shift | |
| Product and mold | |
| Planned output | |
| Actual green output | |
| Rejected units | |
| Curing lot reference | |
| Downtime minutes and cause | |
| Material variance | |
| QC release notes | |

# Appendix E: QC Release Form

No finished product should move from curing to saleable stock without a release record.

| Check | Pass / fail / notes |
| --- | --- |
| Product and lot number | |
| Curing start and release date | |
| Visual defects | |
| Dimensional tolerance | |
| Cube test reference | |
| Breakage during handling | |
| Release decision | |
| QC signature | |

# Appendix F: RMC Delivery Ticket Fields

Ready-mix disputes are best prevented at dispatch. The delivery ticket should align with batch records and site acceptance.

| Field | Purpose |
|---|---|
| Customer and site | confirm delivery responsibility |
| Concrete grade and slump | match order and site requirement |
| Batch time and truck departure | control time-sensitive delivery |
| Water added at plant | trace water-cement discipline |
| Water added on site | allowed only by approved protocol |
| Slump result | site acceptance evidence |
| Customer sign-off | billing and dispute control |

# Appendix G: Board KPI Pack

The board should see a concise operational dashboard, not only accounting profit. The dashboard should connect production, quality, sales, cash and risk.

| KPI | Target direction | Why it matters |
|---|---|---|
| Revenue by product family | up with mix discipline | growth and portfolio health |
| Gross margin by product | stable or improving | pricing and raw material control |
| Reject rate | down | quality and cost leakage |
| On-time delivery | up | customer trust |
| Receivable days | down | cash conversion |
| Inventory age | controlled | curing discipline and stock rotation |
| Lost-time incidents | zero | safety and operational continuity |

# Appendix H: Site Due Diligence Checklist

Land that appears cheap can become expensive if access, drainage, utilities, zoning or ground conditions are poor.

| Item | Required evidence |
|---|---|
| Title / lease | lawyer-reviewed tenure and encumbrance search |
| Zoning | industrial use compatibility and city planning confirmation |
| Access | truck route, turning radius and road condition |
| Geotechnical | bearing capacity, groundwater and earthworks requirement |
| Drainage | stormwater outfall and flood risk |
| Utilities | power route, water source and communications |
| Neighbors | dust, noise, operating hours and social license |

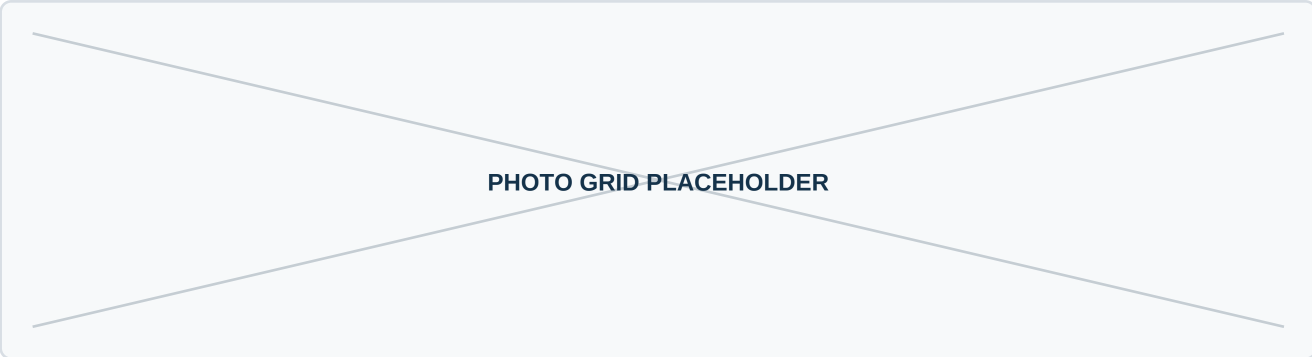
# Appendix I: ESG and Community Commitments

A practical ESG plan should focus on dust, noise, water, washout, road safety, jobs, skills, local sourcing and transparent complaint handling.

| Area | Commitment |
| --- | --- |
| Dust | covered stockpiles, water sprays and housekeeping |
| Water | metering, recycling and compliant discharge |
| Waste | rework broken products where possible; manage cement washout |
| Safety | traffic separation, PPE, training and incident reporting |
| Jobs | local hiring and structured operator training |
| Community | complaint log and response time standard |

# Appendix J: Product Photo Placeholders

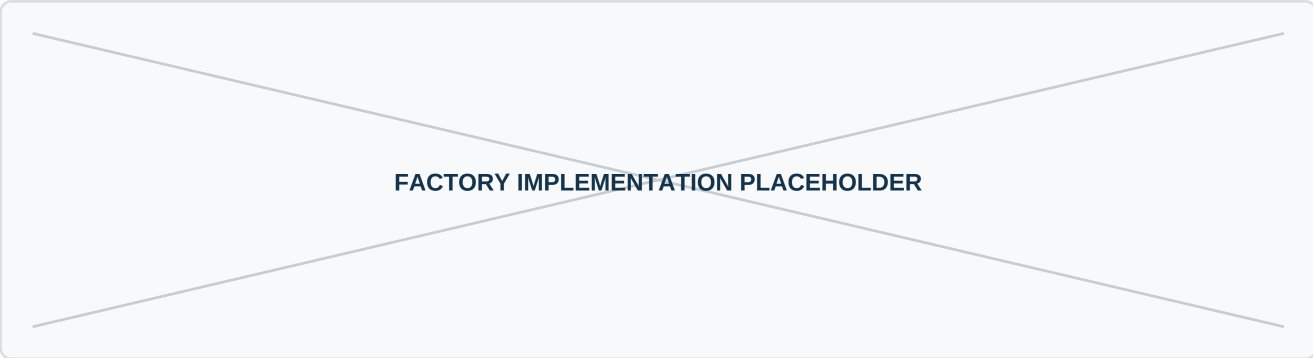
The final investor deck or lender appendix should insert product photos from trial production, not generic stock imagery. Photos should show dimensions, surface finish, packaging, curing and loading.




Insert four to six verified product photos after mold trials.

# Appendix K: Factory Photo Placeholders

During implementation, replace this placeholder with site photos, layout progress, equipment installation and commissioning evidence.




Insert site, equipment, lab and yard photographs.

# Appendix L: Source List

The report uses public sources for context and internally generated assumptions for the financial model. Source links should be refreshed before financing.

| Source | URL | Use in report |
|---|---|---|
| World Bank Uganda overview | https://www.worldbank.org/ext/en/country/uganda | Macro growth, population, industrialization context |
| UBOS 2024 Census portal | https://statistics.ubos.org/nphc/index | Population and household evidence base |
| UBOS publications explorer | https://www.ubos.org/?p_id=126&pagename=explore-publications | Annual GDP, construction input price index and national accounts publications |
| NEMA ESIA process | https://www.nema.go.ug/en/esia/ | Environmental assessment route for project brief, ToR or ESIA |
| URSB business registration | https://ursb.go.ug/services/business-registration/ | Company registration mandate and business formalization benefits |
| UIA investment FAQs | https://ugandainvest.go.ug/contact/faqs/ | Investment licensing and investor facilitation context |
| UIA industrial parks | https://ugandainvest.go.ug/parks/ | Industrial park infrastructure and land facilitation context |
| ERA UEDCL transition | https://www.era.go.ug/event/uedcl-take-over/ | Electricity distribution transition to UEDCL from April 2025 |
| NWSC new connections | https://www.nwsc.co.ug/new-connections/ | Water connection process and supporting documents |
| NBRB concrete guidance | https://nbrb.go.ug/wp-content/uploads/2024/09/GUIDELINES-FOR-THE-DESIGN-AND-CONSTRUCTION-OF-STRUCTURES-WITH-STEEL-CONCRETE-AND-TIMBER.pdf | Concrete water-cement, cube testing, curing and compaction guidance |
| UNBS paver standard webstore | https://webstore.unbs.go.ug/store.php?preview=&src=401 | Precast concrete paving block standard reference |
| Nicomah Uganda | https://nicomah.com/ | Western Uganda construction material and concrete product supplier reference |
| Master Industries Uganda | https://masterindustriesug.com/ | Concrete blocks, pavers, kerbs and culvert competitor reference |
| Tororo Cement | https://www.tororocement.com/ | Cement producer and construction material product reference |
| Hima Cement | https://himacement.co/ | Cement products, blocks and precast reference |
| UIA cement factory news | https://ugandainvest.go.ug/new-300-million-dollar-cement-factory-opening-in-uganda/ | Uganda cement supply landscape context |

# Appendix Worksheet: Demand interviews

This worksheet should be completed during the next diligence phase for demand interviews.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
|---|---|
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |

# Appendix Worksheet: Supplier quotations

This worksheet should be completed during the next diligence phase for supplier quotations.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
| --- | --- |
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |

# Appendix Worksheet: Equipment factory visit

This worksheet should be completed during the next diligence phase for equipment factory visit.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
|---|---|
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |

# Appendix Worksheet: Civil works BoQ

This worksheet should be completed during the next diligence phase for civil works boq.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
| --- | --- |
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |

# Appendix Worksheet: Tax and incentive advice

This worksheet should be completed during the next diligence phase for tax and incentive advice.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
|---|---|
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |

# Appendix Worksheet: Environmental route confirmation

This worksheet should be completed during the next diligence phase for environmental route confirmation.

Convert the findings into action items, model changes, contract conditions or board approvals.

| Question | Evidence / answer |
| --- | --- |
| What decision depends on this workstream? | |
| Who owns the workstream? | |
| What evidence has been collected? | |
| What remains uncertain? | |
| What model assumption changes? | |
| What is the recommended board action? | |